

Electroweak and Higgs Realization from a Premetric Response Atlas

Interaction Reconstruction and Target-Relative Closure

Amos Jay Maley
Independent Researcher

Primitive-standing reference edition
21 September 2026

Abstract

We establish a source-relative realization and response-classification theorem for the one-doublet electroweak–Higgs sector of a premetric bosonic atlas. A common Higgs–gauge source determines the fixation orbit, residual charge direction and rank-three gauge response. The residual action and kinetic form then determine all joint-invariant real gauge supports. Together with the scalar normal quotient, they form a four-atom graded support lattice with one compatible module assignment. Carrier identity, response support and physical particle realization remain distinct.

Beyond the quadratic structure, four invariant response coordinates reconstruct the complete normalized local gauge–Higgs interaction; a fifth is necessary when the target observes the vacuum value. We give an explicit inverse and prove coordinate irredundancy on the stated positive family. The native local classification is its actual source-admitted image, preserving genuine parameter distinctions. More generally, finite polynomial-jet tensors reconstruct the whole declared local law and its differential operations.

For admitted physical channels, full analytic germs determine the meromorphic response on its marked connected domains. An independently bounded rational branch admits a finite determining jet. A single joint source kernel constructs and classifies cyclic state representations, current attachments and bounded continuations; resolvents or complete closed graphs determine unbounded operator domains. These results preserve mixed incoming–outgoing data that marginal comparisons lose. They complement the explicit free-helicity construction and the source-attached photon and pole-transport theorems, whose interacting spectral and asymptotic premises remain independent.

The native operations and their exact incidence maps yield a complete observation and continuation algebra. Whole-interface role occupation has one lawful assignment orbit over each reconstructed response point. It is terminal on branches where the reconstruction data are detectable by the admitted observations; coarser targets retain their explicit native-readout quotient. Constructive deletion results distinguish essential response information from recoverable descriptions, including resolvent-generated channels. The resulting classification strengthens the structural atlas without selecting numerical parameters or introducing additional primitive bosonic carriers.

Contents

1	From a premetric response atlas to electroweak realization	1
2	The common source and the realization domain	2
2.1	The inherited bosonic response atlas	2
2.2	Lawful comparisons and source preservation	3
2.3	Common attachment by source restriction	3
2.4	Source-relative existence	5
3	Licensed one-doublet fixation and the residual carrier	6
3.1	The orbit and its normal direction	7
3.2	Residual charge and the geometric descent map	8
4	Gauge and scalar response of one Higgs carrier	9
4.1	The complete gauge-response form	9
4.2	Canonical coordinates and normalization covariance	10
4.3	The gauge-invariant scalar response	11
4.4	Angular variables and longitudinal content	12
5	Canonical response modules and their occupation	12
5.1	Invariant geometry of the response	12
5.2	Exhaustion of the joint-invariant supports	14
5.3	Role occupation at the module target	15
6	Polarization, photon realization, and resonance authority	16
6.1	Polarization spaces as quotient responses	16
6.2	The free helicity realization	17
6.3	Source attachment and photon asymptotics	18
6.4	Gauge-independent continuation of simple poles	19
6.5	Physical channels and realization domains	20
7	Interactions, process types and physical continuation	20
7.1	Local vertices forced by the doublet kinetic term	21
7.2	A source-evaluated process family	22
8	Terminal comparison and deletion-minimal response support	23
8.1	Complete signatures on each activated domain	23
8.2	Lossless deletion and semantic reconstruction	24
8.3	Deletion-minimality of quadratic amplitude support	24
8.4	Extensions and scope preservation	25
9	The native electroweak observation and continuation algebra	25
9.1	Native objects and source reception	25
9.2	Reconstruction of the complete local law	27

9.3	Joint incidence and fixed-sibling comparisons	28
9.4	The reconstructed algebra and terminal identity	29
10	Reconstruction of the local interaction response	31
10.1	An explicit inverse from response observations	31
10.2	No-repair minimality on the family target	33
11	Reconstruction of physical response and continuation data	33
11.1	Full analytic germs and physical channel reconstruction	34
11.2	A finite determining jet on the rational branch	35
11.3	A joint kernel for states, currents and continuations	35
11.4	Closed operators and their source domains	37
11.5	Mixed attachment and observable kernels	38
11.6	The reconstructed physical interface	39
12	Whole-interface realization and role occupation	40
12.1	Roles fixed by the source work	40
12.2	Reconstruction of mixed incidences	41
12.3	Complete role occupation over each response point	42
12.4	When whole-interface reconstruction is terminal	43
13	Minimal information and reconstructible descriptions	45
14	The electroweak–Higgs realization atlas	47
14.1	Exact branches and their interpretation	49
14.2	Implications for further bosonic realization	50
14.3	Conclusion	51

1 From a premetric response atlas to electroweak realization

Electroweak fixation organizes several gauge and scalar responses around one Higgs carrier. This paper classifies that organization within the licensed one-doublet realization of the premetric bosonic response atlas. A structural carrier fixes the standing identity supporting an admissible response; a physical particle additionally requires propagation, polarization and spectral structure. The Higgs connects these levels through its fixation response without adding a gauge root; its coordinate components remain aspects of one carrier rather than independent carrier identities.

BCRS-I constructs the premetric bosonic response atlas, its complete contextual quotient, the joint gauge-law structure, and the common-source Higgs–electroweak readiness map [1]. Starting from a common source realization, we derive the fixation orbit, residual action and quadratic response modules, reconstruct the local interactions and admitted physical responses, and classify their complete interface work. The central question is when these data determine every response the target can distinguish. Geometry, interaction coefficients and spectral realization contribute different parts of that answer.

The realization domain consists of one complex Higgs doublet, a nonzero fixation branch, an admitted gauge kinetic form and a specified operator order, with all source attachments retained. The local effective action on the doublet is distinguished from the actual global gauge group, its ineffective kernel and spectator sectors. The geometric conclusions hold throughout this branch without numerical coupling values or a numerical vacuum scale. The scalar field content is part of the domain, not a consequence of the resulting mass matrix.

Two mathematical distinctions organize the argument. A lawful coordinate change transports the Higgs representative, gauge generators, kinetic form, current pairing and incident maps together. In addition, the kernel of the gauge-response form belongs to field-coefficient space, whereas the electromagnetic annihilator belongs to the generator algebra. The coupling map connects these spaces and is essential both to the charge interpretation and to normalization covariance.

The principal structural result is a canonical simultaneous decomposition. A nonzero fixation has a one-dimensional connected effective stabilizer, three real orbit-tangent directions and a one-dimensional radial normal. The gauge-action Gram form has rank three. Its null direction, charged two-plane and neutral responsive line, together with the scalar normal line, are uniquely determined by the annihilator, residual representation and orthogonality requirements. The occupation theorem thus follows from the geometry, without assuming a particle count or a physical pole.

The interaction result goes further. A finite invariant chart reconstructs the entire normalized local gauge–Higgs law, including its current couplings and higher vertices. Its image is generally a response family, not a single parameter point. An exact inverse proves completeness, and same-retained-data pairs prove which coordinates cannot be erased. Functional physical responses require a different normal form: complete analytic germs for channels and a joint positive kernel for source-generated states and continuations. Actual resolvents or full closed graphs retain the unbounded domains that a formal differential expression alone would omit.

These reconstructions make the terminal comparison explicit. The native source operations, local-law evaluations, physical response maps and their mixed incidences generate the complete admitted observation algebra. At a fixed reconstructed point, the entire activated interface has a unique compatible role assignment. Its identification with terminal physical information additionally requires the reconstruction data to be recoverable from admitted observations. Where that condition is not met, the native readout quotient supplies the correct coarser identity. Thus uniqueness of a presentation, completeness of an observation family and uniqueness of a realized parameter point are not conflated.

Physical interpretation proceeds through distinct mathematical objects. On an admitted Lorentzian realization, the quadratic equations determine polarization spaces. A stable one-particle sector further requires an admitted positive-energy representation identified with the source response; an unstable

resonance requires an actual physical amplitude or invariant correlator with a justified analytic continuation. The Goldstone–longitudinal amplitude relation applies in its specified high-energy domain, with its finite factors and remainder. Each physical conclusion retains these hypotheses.

Premetric gravity supplies the necessity condition on the nondegenerate comparison arena, and the metric realization enters through the licensed physical-interior interface. Gravity is not another electroweak occupant. The common bosonic realization studied here also remains distinct from the full fermion–boson–gravity physical apex; the fermionic successor retains its own terminal and common-apex domains [2].

2 The common source and the realization domain

2.1 The inherited bosonic response atlas

Write $\mathbb{B}_I(\Gamma)$ for the BCRS-I source-formed architecture at fixed global-form and target data Γ . It contains the source and semantic-use families, complete contextual quotient, qualified charge and weak response packets, one Higgs identity C_H , its order/deformation/fixation mechanisms, and the mixed readiness map $\beta_{H,EW}^{\text{rd}}$. It also retains the actual common assemblies, their intrinsic attachment maps and fixed-input covariance, the complete gauge-law reduct, source-operation networks, and every definedness and failure distinction. A source realization selects one such common assembly A , with its joint attachments rather than independently selected marginal projections.

BCRS-I supplies an exact marked-response rigidity criterion [1, Sec. 14], not a universal equality of responses at different genuine arguments. The realization image classified here retains its actual map from that predecessor and leaves the criterion unchanged.

Definition 2.1 (Common electroweak realization data). *A common realization instance x consists of the following mutually compatible data, with all restrictions and attachment maps retained:*

1. *a source-realized BCRS-I assembly A and its actual mixed Higgs–electroweak readiness point;*
2. *an admitted realization of the inherited gauge and Higgs objects on the declared base, with the actual global group G_Γ , representation, boundary conditions and transition maps;*
3. *a Hermitian rank-two Higgs bundle V , carrying the specified one-doublet effective electroweak action, and a nonzero fixation section on the stated domain;*
4. *the gauge-field coefficient bundle F , its positive kinetic form κ , invariant under the transported residual action as inherited from the admitted gauge kinetic term, the nonzero-coupling map C_g , and the jointly attached current pairing;*
5. *the scalar kinetic and potential data at the declared operator order, together with the precise stationary-branch or Hessian hypotheses used in a claimed excitation theorem;*
6. *any further spacetime, propagation, spectral or asymptotic structure used by a physical projection, recorded separately rather than inferred from the preceding algebraic data.*

Compatibility means that these are data on one instance, with commuting source restrictions, not merely objects whose individual existence has been established. The structural realization domain omits any demand that the additional physical structures in the last item already exist. Their separate domains are defined when used.

The definition specifies the common realization domain. Its local algebraic part is nonempty, as the explicit doublet family below shows. Attachment of that family to a particular premetric source additionally requires the actual realization maps and common-domain hypotheses. Those maps identify

the chosen field H with the source; algebraic consistency alone is not an existence theorem for every attachment.

The one-doublet branch and its coupling conventions are supported by the electroweak symmetry source [8, Defs. 2.1–2.2 and Lem. 2.4]. The common Higgs identity and its permitted metric envelope are inherited through the Higgs bridge source [5]. SYM-8’s perturbative spectral bundle belongs to a further physical extension, whose separate hypotheses are required wherever that extension is used.

2.2 Lawful comparisons and source preservation

Definition 2.2 (Comparison of common instances). *A lawful comparison is an invertible comparison of the declared realization data carrying all source restrictions, actual attachments, global and boundary data, metrics, operators and selected branches together. At a fixed base it includes the properly typed changes of local frame and presentation. A comparison of source records which is not invertible on representatives is used only through its already proved intrinsic descent map; it is not assigned a fictitious inverse. Physical comparisons additionally preserve the independent authority used by the physical assertion.*

Proposition 2.3 (Compatibility under composition). *Common instances and their lawful invertible comparisons form a groupoid. Forgetting the realization data gives a functor to the corresponding BCRS-I intrinsic common-assembly groupoid. Every construction commuting with these comparisons is defined independently of its local presentation, while retaining its actual source instance.*

Proof. Identities preserve all structural maps. Composites preserve them because the corresponding diagrams compose, and inverse diagrams preserve them in the opposite direction. Source restriction commutes with identities and composition, so it is functorial. A comparison-covariant construction has equivalent values on two representatives of one instance. This does not identify different instances or suppress an attachment for which no comparison exists. $\square$

Local and global data play different roles. The effective doublet action may be represented by $U(2)$, while the stabilizer in G_Γ is the inverse image of its effective stabilizer. A nowhere-zero section gives local stabilizer reductions that glue by the retained transition functions, without requiring a global unitary gauge. Boundary and asymptotic transformations remain separate from the proper local quotient. Their physical charges require the corresponding boundary structure in addition to an infinitesimal stabilizer.

Proposition 2.4 (Conservative realization). *The electroweak realization construction does not alter a BCRS-I source, complete signature, carrier identity or mechanism fibre. Equality in a coarser realization image does not imply equality in the predecessor.*

Proof. All constructions take a retained predecessor object as an argument. Their lawful comparisons commute with its restriction maps. The predecessor equivalence relation is therefore unchanged. A map may identify distinct inputs in its image, but recovering equality of the inputs would require injectivity or a faithful reconstruction theorem. Neither is inferred from the map’s existence. This is the exact conservative successor rule of BCRS-I [1, Thm. 14.4]. $\square$

2.3 Common attachment by source restriction

A common attachment can be obtained directly from one electroweak source realization. Restrict its native source data to form a BCRS-I assembly, then attach the original realization to that assembly through their shared source maps. On the declared source-admitted branch, this gives the common

attachment without an additional BCRS-specific matching condition. The result is source-relative; it does not assert a lift of every premetric instance.

Definition 2.5 (Dependency-expanded electroweak source branch). *Let $\mathfrak{U}_{EW}^{1H}$ be the independently admitted one-doublet source branch with the following native source data retained:*

1. *the instantiated physical-interior and Standard Model gauge construction, including the actual A1–A6, PSM1–PSM6 and SR1–SR7 antecedents, the common boundary and global-form data, and the unmodified color and other spectator source records;*
2. *the Higgs source’s bridge-witness package in that same construction, with its order and deformation quotients, reduced mediation map, residual normal form, and gauge–vacuum compatibility;*
3. *the one-doublet structural realization used in the local SYM-8 construction: its admitted metric and differential setting, inherited representation and current map, nonzero fixation branch, kinetic forms, nonzero couplings, and the actual proper-gauge and reduction maps.*

The dependency expansion means retaining the witnesses used by the native source construction, rather than recovering them from the displayed coarse Standard Model tuple. No BCRS term, mixed-readiness point, assembly, response equality, or role-occupancy conclusion occurs in this source specification. Spectral or asymptotic certificates are retained only when a physical extension actually supplies them.

The source dependencies occur within one common construction. On the matter-bearing Standard Model source, the primitive-generation record is the original-duty interpretation and complete occupation retained by BCRS-I [1, Def. 12.18]. The common attachment transports that record with the actual matter and boundary incidences. The scalar normal dimension, gauge-response rank and physical spectral conditions below are determined by their electroweak–Higgs constructions; none is inferred from the generation count. The standalone Standard Model theorem carries the realized-interior, gauge, representation, charge, anomaly, vacuum and bridge dependency chain [3, Thms. 7.2, 9.1–9.3 and 9.7]. The full gauge-carrier source retains its Higgs and spectator dependencies in one electroweak structural tuple [10, Sec. 18.2 and App. A1]. SYM-8 receives the Standard Model tuple, gauge–vacuum reduction, residual normal form and Higgs endpoint as joint interfaces of its selected realization [8, Sec. 1.2, Def. 2.1 and Apps. D.1–D.2]. Its local bridge map is computed on that inherited representation and fixation class [8, Sec. 11.2 and Thm. 11.1].

The coordinates H , g_1 , g_2 , and v alone specify only part of the data required by this branch. The complete transport, loop, boundary, Higgs and color records must also be retained; a coarse gauge-group isomorphism cannot recover them. A source supplied only on its electroweak face yields an attachment on that face. The full assembly additionally requires the spectator source attachments.

Lemma 2.6 (Native-source shadow). *Every $u \in \mathfrak{U}_{EW}^{1H}$ determines a source-realized electroweak instance $\Xi(u)$ of BCRS-I. Its Higgs, charge, weak, and mixed-incidence restrictions are the restrictions already present in u . No comparison of independently chosen marginal sources is needed.*

Proof. Retain the original common gauge/Higgs construction underlying u , and forget only the additional realization outputs not used at the BCRS-I target. The physical-interior antecedent remains attached; forgetting a metric output does not erase the premise under which its source was admitted. The full gauge source retains the A1–A6, PSM1–PSM6 and SR1–SR7 maps, including the original SR7 survivor projection. The full Higgs witness remains attached to this very source, with its actual mediation map. The spectator sources are not changed. These are precisely the native data in BCRS-I’s source-realized electroweak instance [1, Sec. 10].

In detail, the charge restriction uses the actual representation and normalized charge witnesses of the common gauge source; the weak restriction uses its witnessed weak-chiral action. The Higgs

restriction uses the quotient-reduced bridge witnesses, not merely their common root name. The mixed restriction is the original order-to-deformation mediation graph together with these three pointed ports and every fixed ambient input. All boundary, continuation, loop, guard and failure coordinates are restrictions of the same record. Neither the newly computed stabilizer nor a physical spectrum value is used to form these source restrictions. $\square$

Theorem 2.7 (Source-relative common-attachment construction). *For every $u \in \mathfrak{U}_{EW}^{1H}$, there is a constructed BCRS-I common assembly A_u , its mixed-readiness point d_u , and a common electroweak realization x_u whose predecessor is A_u and whose structural realization is the one already carried by u . The attachment maps preserve the full fine input packets, actual mixed incidence, boundary, continuation, domains and failure records. The construction is natural under lawful native source comparisons.*

Proof. Apply BCRS-I's all-instance source reception to $\Xi(u)$ from Lemma 2.6. It gives certified source terms for the actual Higgs, charge and weak restrictions. Their complete contextual quotients form the fine packets; their source-preserved root incidences certify activity. For the mixed term, receive the retained mediation graph with its exact input ports and fixed ambient data. The source-relative mixed-readiness theorem therefore constructs $d_u = (h_u, x_{Q,u}, x_{wk,u}, [\lambda_u])$. The simultaneous source-realization theorem constructs A_u with the same source restrictions and retained spectator data [1, Secs. 10, 12 and 14]. This invokes the accepted constructive source theorem, not its conditional marked-response singleton criterion.

Now retain the structural realization component of u . Its source ports are the very ports just received by BCRS-I. On the common native record the comparison maps are identities. In a different source presentation they are the certified comparison arrows carried by u , and their squares commute because both routes restrict that same record. Thus the attachment does not add an equation between two separately selected physical inputs. It records the equality of two restrictions of one input.

The actual gauge–vacuum and residual source maps remain attached, so the construction does not infer an admissible reduction from a vev coordinate. The one-doublet geometry and kinetic map are supplied by the native realization, rather than deduced from source readiness. Consequently all clauses of the common-instance definition are satisfied, producing x_u . Finally, source reception, quotient formation and the native realization maps commute with the admitted source comparisons. The fixed-input domain-reflection theorem of BCRS-I preserves the exact mixed instance along those arrows [1, Sec. 11]. The joint attachment construction is therefore natural. $\square$

2.4 Source-relative existence

Let $\mathcal{J}$ denote the groupoid of the complete native pointed source records used above, with all original source antecedents and attachment maps retained. Let $\mathcal{A}_I$ denote the BCRS-I source-realized assembly groupoid. Because the predecessor quotients can forget presentation data, an assembly need not determine its original full native source record. The comparison therefore uses the *source-decorated* assembly groupoid $\tilde{\mathcal{A}}_I$. Its objects are triples $(j, A, \mathfrak{r})$, where $j \in \mathcal{J}$, $A \in \mathcal{A}_I$, and $\mathfrak{r}$ is a retained witness of the accepted source reception and simultaneous assembly of j into A . The witness retains the source terms together with their actual quotient and attachment maps. Its arrows are lawful native-source and assembly comparisons whose squares with these reception maps commute. Identities, composites and inverses inherit those commuting squares, so this is a groupoid.

The retained source defines the projection $\sigma(j, A, \mathfrak{r}) = j$, and a separate forgetful functor $(j, A, \mathfrak{r}) \mapsto A$ to the accepted predecessor. The first map projects supplied data rather than reconstructing it from the bare assembly; the second leaves the predecessor unchanged. Write $j(u)$ for the native source record retained by $u \in \mathfrak{U}_{EW}^{1H}$. The comparison object is the groupoid pullback

$$\mathcal{X}_{EW} = \mathfrak{U}_{EW}^{1H} \times_{\tilde{\mathcal{J}}}^{\sim} \tilde{\mathcal{A}}_I.$$

An object is $(u, (j', A, \mathfrak{r}), \theta)$, where $\theta : j(u) \simeq j'$ is a lawful comparison of the full joint source records, not merely an isomorphism of carrier names.

Corollary 2.8 (Canonical source-relative section). *The construction in Theorem 2.7 defines a section s of the source projection p :*

$$s(u) = (u, (j(u), A_u, \mathfrak{r}_u), \text{id}_{j(u)}),$$

$$p : \mathcal{X}_{EW} \longrightarrow \mathfrak{U}_{EW}^{1H}, \quad p \circ s = \text{id}.$$

Every independently supplied native source instance therefore gives an inhabitant of the exact common-attachment fibre. There is no remaining BCRS-specific attachment-existence condition on this source image.

Proof. The theorem constructs A_u by reception of $j(u)$. Retain that construction, with its actual source terms and commuting quotient and attachment maps, as $\mathfrak{r}_u$. Thus $(j(u), A_u, \mathfrak{r}_u)$ is a source-decorated assembly whose source projection is exactly $j(u)$. The identity of this retained record supplies the pullback comparison. On a lawful native-source arrow, the theorem's natural reception supplies the corresponding assembly arrow and commuting reception squares. These define s on arrows and preserve identities and compositions. Forgetting the decorated assembly and comparison returns precisely the original source object and arrow, proving $p \circ s = \text{id}$. In particular, choosing any supplied native source instance gives a common-attachment witness; no inverse of a predecessor quotient and no separate search for equal marginal outputs is required. $\square$

The section establishes a common attachment for every instance of the source-admitted branch, including instances for which physical pole or asymptotic fields have not been supplied. It is not a realization theorem for every formal BCRS-I source or every global-form fibre, and it does not supply a missing physical spectral bundle.

The local algebraic branch admits an explicit nonempty parameter family. Take positive r, g_1, g_2, c , the standard doublet action, canonical positive kinetic forms, $\phi = (0, r)^T$, and $V(H) = c(H^\dagger H - r^2)^2$. This gives a nonzero stationary branch and a positive radial Hessian by the calculation below, without empirical calibration. This algebraic example and native-source admission establish different facts. The former exhibits solutions of the nondegenerate one-doublet realization equations; the latter retains standing, the physical-interior antecedent and the complete source attachments required by Definition 2.5. The numerical family alone supplies none of those additional witnesses.

3 Licensed one-doublet fixation and the residual carrier

The inherited Standard Model load specifies the one-doublet realization branch. The structural spine gives its conventional presentation as $H \sim (1, 2)_{1/2}$, while retaining the potential and its parameters as downstream data [3, Def. 1.7 and Status 1.11]. The Higgs endpoint identifies the one bridge/fixation carrier C_H and its order, deformation, and fixation roles [4, Def. 7.3 and Thms. 7.4, 8.4]. Selection of a physical vacuum and metric belongs to the realization data. The metric-envelope theorem places the scale-bearing use of this carrier within a licensed comparison and transport regime [5, Defs. 5.1, 7.1 and Thm. 8.3]. The realization is accordingly attached to the exact common Higgs–electroweak source datum, with its joint restrictions intact.

Definition 3.1 (Attached one-doublet realization). *An attached one-doublet realization over a BCRS-I mixed-readiness datum $d = (h, x_Q, x_{\text{wk}}, [\lambda])$ consists of the following data on one common continuation domain:*

1. a licensed spacetime comparison and differential structure, with its actual boundary and continuation restrictions;
2. a Hermitian rank-two complex Higgs bundle E , the inherited electroweak representation, and its covariant derivative;
3. a selected smooth nonvanishing fixation section ϕ , considered only modulo the admitted proper gauge transformations;
4. the nonzero coupling map C_g and positive gauge kinetic form κ , invariant under the transported residual action and transported together with the matter-current pairing;
5. the actual gauge–vacuum compatibility and admissible-reduction witnesses, including the residual transition, transport, loop, representation, charge, anomaly, deformation, and restoration records;
6. a source attachment whose Higgs, charge, weak, and mixed restrictions are precisely those of d , preserving the fixed ambient input data.

No component of this definition asserts a pole, a particle count, or uniqueness of all admissible marked responses. The complete realization domain $\mathcal{D}_{\text{geom}}^{1H}$ is the family of these attached witnesses modulo their lawful presentation arrows.

Gauge–vacuum compatibility requires neutrality of every admitted compatibility probe, and admissible reduction preserves the complete prior ledgers [6, Defs. 5.4–5.10 and Thms. 10.1, 12.1]. The residual record consists of the induced witness ledger, including its maps and compatibility data [7, Defs. 4.1–4.3 and Thm. 15.1]. These requirements distinguish an attached point of $\mathcal{D}_{\text{geom}}^{1H}$ from an unattached doublet model or a subgroup specified only by name.

3.1 The orbit and its normal direction

The following geometry is fibrewise: it applies in every admitted local trivialization and agrees on overlaps, with global sectors and boundary conditions held fixed. Write $E_x \simeq \mathbb{C}^2$, let $r^2 = \langle \phi, \phi \rangle > 0$, and use the effective unitary action on this doublet. Its effective image is $U(2)$. The faithful global gauge group on the full matter and extended-operator system remains separate data.

Theorem 3.2 (One-doublet orbit and connected stabilizer). *At every nonzero fixation vector, the effective unitary orbit is the radius- r three-sphere. Its stabilizer is conjugate, in the effective doublet action, to $\{\text{diag}(e^{i\theta}, 1)\}$. The connected electroweak annihilator is consequently one-dimensional. With Hermitian generators $T_a = \sigma_a/2$, $Y\phi = y_H\phi$, and $y_H \neq 0$, it is*

$$\mathfrak{h}_\phi = \mathbb{R}Q, \quad Q = T_3 + \frac{Y}{2y_H},$$

in a frame in which $\phi = (0, r)^T$.

Proof. Every unit vector can be extended to a unitary frame. Sending one such frame to another proves that $U(2)$ is transitive on the sphere, and unitarity preserves the radius. A unitary matrix fixing $(0, r)^T$ has second column $(0, 1)^T$; orthonormality forces its first column to be $(e^{i\theta}, 0)^T$. This proves the effective stabilizer claim. For the electroweak generator $X = a_1T_1 + a_2T_2 + a_3T_3 + bY$,

$$X\phi = \frac{r}{2} \begin{pmatrix} a_1 - ia_2 \\ -a_3 + 2by_H \end{pmatrix}.$$

Thus $X\phi = 0$ exactly when $a_1 = a_2 = 0$ and $a_3 = 2by_H$, giving the displayed line. Equivariance conjugates the annihilator when the frame changes. These are the explicit local claims of SYM-8's connected-stabilizer theorem [8, Thm. 5.1]; no residual generator is assumed in the target. $\square$

Theorem 3.3 (Angular and radial decomposition). *The derivative of the unitary action is $a_\phi(X) = iX\phi$, for Hermitian X . Its image and the real orthogonal decomposition of the doublet are*

$$\begin{aligned} \text{im } a_\phi &= \{\eta : \text{Re}\langle\phi, \eta\rangle = 0\} = i\mathbb{R}\phi \oplus \phi^{\perp c}, \\ E_{x\mathbb{R}} &= \mathbb{R}\phi \oplus i\mathbb{R}\phi \oplus \phi^{\perp c}. \end{aligned}$$

Hence the tangent quotient by the angular gauge directions is one real dimension, represented by the radial normal line $\mathbb{R}\phi$.

Proof. Since X is Hermitian, $\langle\phi, X\phi\rangle$ is real, so every $iX\phi$ is tangent to the sphere. In the frame $\phi = (0, r)^T$, a Hermitian matrix has the form $X = \begin{pmatrix} a & z \\ \bar{z} & b \end{pmatrix}$, with a, b real. Its action gives $iX\phi = (irz, irb)^T$. The arbitrary complex z supplies $\phi^{\perp c}$, while real b supplies $i\mathbb{R}\phi$. These exhaust the three-dimensional real tangent space. The radial line is its real orthogonal complement. $\square$

For a nonvanishing section these spaces form smooth bundles of constant rank. In the real inner product $(u, w)_\mathbb{R} = \text{Re}\langle u, w\rangle$, their projectors are

$$P_{\text{rad}}(\eta) = \frac{\phi(\phi, \eta)_\mathbb{R}}{r^2}, \quad P_{\text{phase}}(\eta) = \frac{i\phi(i\phi, \eta)_\mathbb{R}}{r^2}, \quad P_\perp = I - P_{\text{rad}} - P_{\text{phase}}.$$

They are smooth and obey $P_{U\phi} = UP_\phi U^{-1}$ under every actual unitary transition map. These real projectors distinguish the radial and phase lines, which the complex line projector alone cannot separate. They define the bundles without a single global unitary-gauge chart. The constant-rank statement excludes zeros of ϕ and changes of boundary conditions or global reduction data.

3.2 Residual charge and the geometric descent map

In the convention $y_H = 1/2$, the derived generator is $Q = T_3 + Y$. Applying its inherited representation to the left-quark doublet gives $(2/3, -1/3)$; to the left-lepton doublet it gives $(0, -1)$; and the charged right-handed singlets retain $2/3, -1/3, -1$. These values follow by evaluating the inherited hypercharge and representation ledger, without an independent charge fit. Furthermore $[Q, T_\pm] = \pm T_\pm$, so the charged weak directions have relative charges ± 1 [8, Props. 7.1 and Thm. 7.2; Eq. (17)].

Definition 3.4 (Residual electromagnetic structural carrier). *For an attached realization, C_γ^{str} denotes the induced residual charge/transport record consisting of the derived connected line $\mathfrak{h}_\phi$, its inherited matter-current pairing, and the actual residual transition, transport, loop, charge, and anomaly data. It is the electromagnetic specialization of the inherited residual carrier, not an additional premetric root and not yet a physical photon one-particle sector.*

Theorem 3.5 (Attached geometric electroweak descent). *On $\mathcal{D}_{\text{geom}}^{1H}$, the assignment*

$$\beta_{H,EW}^{\text{geom}} : \mathcal{D}_{\text{geom}}^{1H} \longrightarrow \mathcal{P}_{\text{geom}}^{1H}$$

is well defined when its output records the fixation class, connected stabilizer, angular and radial bundles, induced response form, C_γ^{str} , mixed incidence, and the inherited continuation and failure records. It commutes with every admitted same-target restriction. Changing one input presentation with its incident maps transported does not change the intrinsic output while the other intrinsic inputs are held fixed.

Proof. The fixation class is already a proper-gauge quotient. The annihilator and tangent construction are equivariant: for a unitary source arrow U , $a_{U\phi}(UXU^{-1}) = Ua_\phi(X)$. The Hermitian form transports the orthogonal decomposition, and the response form is the Gram form of this same action with its coupling map. The inherited current pairing and residual ledgers travel through the same source

reduction; none is recomputed from an independently selected representative. Each constituent therefore defines a map on the attached diagram and commutes with restriction. Their joint record does too. The full mixed attachment ensures that a change in one presentation changes its incident coordinate maps, not an intrinsic sibling input. This is the fixed-input comparison established at the BCRS-I handoff, now applied to equivariant realization operations. A change of an actual coupling, branch, attachment, boundary class, or inherited holonomy is not among these presentation arrows. $\square$

The geometric map gives the structural part of licensed electroweak descent. A physical extension retains this record and adds spectral fields only on the domain where their independent hypotheses hold. The structural result remains valid when those physical fields are unresolved.

4 Gauge and scalar response of one Higgs carrier

4.1 The complete gauge-response form

Let F be the real space of gauge-field coefficients and $C_g : F \rightarrow \mathfrak{g}_{EW}$ its nonzero-coupling isomorphism. This map relates the distinct generator and field-coefficient spaces: the null-field coefficients map to the generator annihilating the Higgs. At an attached fixation section, the action and its response form are

$$L_\phi(u) = i\rho(C_g u)\phi, \quad \mu_\phi(u, w) = 2 \operatorname{Re}\langle L_\phi(u), L_\phi(w) \rangle.$$

The factor two corresponds to writing the quadratic response term as $\frac{1}{2}\mu_\phi(A, A)$. The factor i gives the tangent action and cancels out of the Gram form.

Theorem 4.1 (Rank-three response and exact residual kernel). *The Higgs-induced form is positive semidefinite, and*

$$\ker \mu_\phi = \ker L_\phi = C_g^{-1}(\mathfrak{h}_\phi), \quad \operatorname{rank} \mu_\phi = 3.$$

Its quotient form on $F/\ker \mu_\phi$ is positive definite. The induced map

$$\bar{L}_\phi : F/\ker \mu_\phi \longrightarrow T_\phi(U(2)\phi)$$

is an isomorphism, with quotient metric $\mu_\phi/2$ equal to the pullback of the real Higgs metric.

Proof. The diagonal value is $\mu_\phi(u, u) = 2\|L_\phi(u)\|^2$, so it vanishes exactly when $C_g u$ annihilates ϕ . Theorem 3.2 gives a one-dimensional annihilator, and invertibility of C_g preserves that dimension. Rank-nullity on the four-dimensional coefficient space gives rank three. The descended form is positive definite because all and only its zero directions were divided out. The image is the full orbit tangent space by Theorem 3.3; the first isomorphism theorem and the defining Gram identity prove the remaining claims. The kernel and rank agree with the independently established SYM-8 response theorem [8, Lem. 6.1 and Thm. 6.2]. $\square$

Theorem 4.2 (Charged and neutral response decomposition). *With the actual positive gauge kinetic form κ , invariant under the residual action transported to F by C_g , the response operator $M = \kappa^{-1}\mu_\phi$ has a two-dimensional charged real subspace, one neutral null line, and one neutral positive-response line. These are invariantly defined together with the residual charge action. Complexifying the charged real subspace gives the two conjugate charge lines conventionally denoted W^+ and W^- .*

Proof. At $\phi = (0, r)^T$, the adjoint action of the stabilizer rotates $\operatorname{span}_{\mathbb{R}}(T_1, T_2)$ and fixes $\operatorname{span}_{\mathbb{R}}(T_3, Y)$. The first space is the nontrivial real rotation representation; the second is a trivial two-dimensional representation. There is no nonzero invariant real linear map between them: an invariant functional on

a plane rotated through all angles must vanish. Thus both invariant symmetric forms κ and μ_ϕ are block diagonal, and their restrictions to the charged plane are scalar multiples of its rotation metric. The charged Gram form is positive, since that plane contains no annihilator. In the neutral plane the kernel has dimension one and the rank is one. Its κ -orthogonal complement is therefore the unique positive eigenline of the neutral self-adjoint operator. The commutators $[Q, T_\pm] = \pm T_\pm$ identify the two complex charge lines. The decomposition uses the joint charge and response structure, so it remains meaningful even if an unrelated numerical eigenvalue coincidence occurs. $\square$

4.2 Canonical coordinates and normalization covariance

For the canonically normalized kinetic form, take $\phi = (0, v)^T/\sqrt{2}$, $y_H = 1/2$, and positive nonzero couplings g_2, g_1 . Common sign conventions can be absorbed into the generator and current coordinates. Direct substitution gives

$$\mu_\phi = \frac{v^2}{4} \begin{pmatrix} g_2^2 & 0 & 0 & 0 \\ 0 & g_2^2 & 0 & 0 \\ 0 & 0 & g_2^2 & -g_2 g_1 \\ 0 & 0 & -g_2 g_1 & g_1^2 \end{pmatrix}.$$

Consequently

$$\begin{aligned} s_W &= \frac{g_1}{\sqrt{g_1^2 + g_2^2}}, & c_W &= \frac{g_2}{\sqrt{g_1^2 + g_2^2}}, \\ A &= s_W W^3 + c_W B, & Z &= c_W W^3 - s_W B, \\ m_{A,\text{quad}}^2 &= 0, & m_{W,\text{quad}}^2 &= \frac{g_2^2 v^2}{4}, & m_{Z,\text{quad}}^2 &= \frac{(g_2^2 + g_1^2) v^2}{4}. \end{aligned}$$

Here A and Z are coordinates on the derived response lines. The subscript “quad” distinguishes the coefficients of the quadratic response from interacting pole positions. The current pairing is

$$g_2 T_3 W^3 + g_1 Y B = e Q A + \frac{g_2}{c_W} (T_3 - s_W^2 Q) Z, \quad e = g_2 s_W = g_1 c_W.$$

The interaction map thus relates the residual null field directly to the inherited charge generator [8, Thm. 6.3 and Prop. 7.1].

Proposition 4.3 (Full response covariance). *If coefficient coordinates change by $A' = N A$, the lawful transported data are*

$$\kappa' = N^{-T} \kappa N^{-1}, \quad \mu' = N^{-T} \mu N^{-1}, \quad J' = J N^{-1}, \quad C'_g = C_g N^{-1}.$$

Then $M' = N M N^{-1}$. Rank, kernel, generalized eigenvalues, charge action, and current pairings are preserved. Simultaneously rescaling $Y' = c Y$, $g'_1 = g_1/c$, and $y'_H = c y_H$, for $c \neq 0$, also preserves the representation-valued coupling map and the derived stabilizer line.

Proof. Substitute $A = N^{-1} A'$ into each quadratic and linear expression. Inverting κ' gives $\kappa'^{-1} \mu' = N \kappa^{-1} \mu N^{-1}$. The coupling and current maps make the same substitution, preserving their evaluations. Hypercharge rescaling preserves $g_1 Y$ and $Y/(2y_H)$ identically. These calculations extend the full neutral-packet covariance in SYM-8 to the complete coefficient space [8, Lem. 2.4 and Prop. 6.4]. $\square$

4.3 The gauge-invariant scalar response

Theorem 4.4 (Radial quotient and invariant scalar witness). *The map $s(H) = H^\dagger H$ separates the effective unitary orbits of a doublet. Near $\phi \neq 0$, it is a smooth coordinate on the one-dimensional orbit space, and*

$$ds_\phi(\eta) = 2 \operatorname{Re}\langle \phi, \eta \rangle, \quad \ker ds_\phi = T_\phi(U(2)\phi).$$

Therefore the radial normal defines one nonzero gauge-invariant scalar response direction without adding a second Higgs carrier.

Proof. Unitary transformations preserve s , and vectors with the same nonzero value of s lie on the same orbit by Theorem 3.2. Zero is the sole orbit at $s = 0$. Differentiation gives the displayed formula; it is nonzero on the radial line and has exactly the orbit tangent as kernel. The radial coordinate $r = \sqrt{s}$ is smooth for $s > 0$. The quotient response is consequently one-dimensional and witnessed by an invariant function of the already attached doublet. This is not a claim about a pole of the operator-valued correlator of $H^\dagger H$. $\square$

Theorem 4.5 (Renormalizable scalar normal form). *Every real gauge-invariant nonderivative polynomial of total field degree at most four for one doublet has the form*

$$V(H) = a + b s(H) + c s(H)^2.$$

For a nonzero stationary branch $s_0 = r^2$, stationarity is $b + 2cs_0 = 0$. If its radial Hessian is positive and nondegenerate, then $c > 0$, $b < 0$, and $s_0 = -b/(2c)$. For these fixed coefficients the minima form exactly one effective unitary orbit.

Proof. Invariance and orbit transitivity imply that the polynomial is determined by its restriction to $(0, t)^T$, with real t . The transformation $t \mapsto -t$ is unitary, so this restriction is even. Its degree is at most four and it therefore equals $a + bt^2 + ct^4$. Every vector has the same value as one with $t = \sqrt{s}$, proving the formula on the entire doublet space. At a nonzero vector, $dV = (b + 2cs) ds$, and $ds \neq 0$, giving the stationarity equation. The radial Hessian has the sign of c ; the exact formula is proved below. Hence strict radial positivity forces $c > 0$, and $s_0 > 0$ then forces $b < 0$. Completing the square gives $V = a - b^2/(4c) + c(s - s_0)^2$, whose minima are precisely the radius- $\sqrt{s_0}$ sphere, one orbit. $\square$

This normal form applies to the declared local renormalizable polynomial target, with its coefficients left as parameters. Effective quantum potentials, thermal potentials and richer scalar targets lie outside this conclusion.

Theorem 4.6 (Hessian descent and scalar response rank). *More generally, let $V = F(s)$ with F twice differentiable near $s_0 > 0$. On a stationary nonzero branch, $F'(s_0) = 0$, and*

$$\operatorname{Hess}_\phi V(\eta, \zeta) = 4F''(s_0) \operatorname{Re}\langle \phi, \eta \rangle \operatorname{Re}\langle \phi, \zeta \rangle.$$

The angular gauge tangent lies in its kernel. If $F''(s_0) > 0$, the Hessian descends to a positive, rank-one scalar response on the radial quotient. In canonical coordinates $\phi = (0, v)^T/\sqrt{2}$ and $H = (0, v + h)^T/\sqrt{2}$, its scalar quadratic coefficient is $m_{h, \text{quad}}^2 = v^2 F''(v^2/2)$.

Proof. The chain rule gives $\operatorname{Hess} V = F''(s) ds \otimes ds + F'(s) \operatorname{Hess} s$. The second term vanishes at stationarity. Substituting the derivative from Theorem 4.4 gives the formula. Its only nonzero direction is radial when $F''(s_0) \neq 0$, and its sign is that of $F''(s_0)$. Finally $s(h) = (v + h)^2/2$, so $ds/dh = v$ at zero, giving the stated coefficient for the canonically normalized real field. No particle-pole theorem enters this computation. $\square$

At $F''(s_0) = 0$ the radial quadratic response is degenerate; for $F''(s_0) < 0$ it is unstable at this order. The carrier identity C_H is common to these branches, but supplies no stable Higgs resonance on a degenerate or unstable branch.

4.4 Angular variables and longitudinal content

Theorem 4.7 (Structural angular nonmultiplication). *The three angular directions of the attached doublet and the three nonzero gauge-response directions are related by the isomorphism $\bar{L}_\phi$ of Theorem 4.1. Changing the angular chart, or using an invertible custodial repackaging of the same doublet, adds no independent scalar quotient direction and no new Higgs carrier.*

Proof. The source and target of $\bar{L}_\phi$ are the quotient gauge coefficients and the orbit tangent, respectively. They describe one three-dimensional geometric incidence. The orbit tangent is removed in the scalar quotient, leaving the radial direction by Theorem 4.4. An invertible chart or doublet repackaging changes neither space nor the quotient map. A purported additional scalar quotient direction would require a tangent class outside the radial line, contradicting the explicit decomposition. The identity of C_H remains that of the inherited bridge carrier. $\square$

The angular isomorphism is an exact geometric relation. The dynamical Goldstone–longitudinal theorem instead relates amplitudes. For an attached perturbative BRST and amplitude realization satisfying the SYM-8 translation hypotheses, the latter has the form

$$\mathcal{A}(V_L^{a_1}, \dots, V_L^{a_n}; X) = \left(\prod_{k=1}^n C_{a_k} \right) \mathcal{A}(\pi^{a_1}, \dots, \pi^{a_n}; X) + \mathcal{R}_{m/E}, \quad \mathcal{R}_{m/E} = O(m/E),$$

with the admitted external-state or resonance treatment, kinematic regime, finite normalization factors, and remainder all retained [8, Prop. 10.1 and Thm. 10.2]. The estimate for $\mathcal{R}_{m/E}$ comes from the amplitude hypotheses, not from the geometric isomorphism. The high-energy replacement is an amplitude relation within this domain, neither an additional-particle assertion nor an exact all-energy field identity. This is the regime-scoped sense of the tensor-content conclusion in the earlier L1 equivalence paper [9, §§3–6].

5 Canonical response modules and their occupation

The response form, residual action and scalar normal geometry determine canonical subspaces for the distinct electroweak mechanisms. Their joint invariance yields an exhaustive support classification and a unique role occupation at the quadratic realization target. The classification concerns this geometric realization; responses to different genuine arguments of the full premetric law remain distinct when their standing signatures distinguish them.

5.1 Invariant geometry of the response

Fix one common realization instance. Let V be its Hermitian Higgs doublet, regarded as a real inner-product space when taking orthogonal complements, and let $\phi \neq 0$ be a local representative of its fixation section. Let F be the real four-dimensional electroweak field-coefficient space with positive kinetic inner product κ . Its actual coupling map is $C_g : F \rightarrow \mathfrak{g}_{\text{EW}}$. The linearized action is

$$K_\phi(a) = d\rho(C_g a)\phi, \quad M_\phi = 2K_\phi^* K_\phi.$$

In the Hermitian-generator convention of the preceding section, $d\rho(X) = i\rho(X)$, so K_ϕ is the same map as L_ϕ . The adjoint uses both retained metrics; the factor two is the convention for a quadratic mass term written with coefficient one half. Overall normalization does not affect any subspace below. The metric and coupling map transform along with their operands under every admitted change of frame. In particular, the generator annihilator and the null field direction are related by C_g , not identified literally.

The effective connected stabilizer of ϕ acts on F by transport of its adjoint action through C_g . This action is orthogonal for κ . Write it as R_ϕ ; the actual stabilizer in the retained global gauge group is its inverse image under the effective representation. The spectator, kernel and discrete global data remain part of the target. The preceding one-doublet calculation establishes

$$\dim \ker K_\phi = 1, \quad \dim \operatorname{im} K_\phi = 3, \quad V_{\mathbb{R}} = T_\phi(G\phi) \perp \mathbb{R}\phi.$$

Here and below the statements are fibrewise; transition maps are retained when the spaces occur in bundles.

Definition 5.1 (Residual fixed and moving subspaces). *The fixed subspace and its orthogonal complement are*

$$F^{R_\phi} = \{a : R_\phi(t)a = a \text{ for every admitted } t\}, \quad F_{\text{mov}} = (F^{R_\phi})^\perp.$$

Equivalently, the fixed-space projector is the average of the compact effective residual action. This characterization does not select a phase or a basis in the charged plane.

Lemma 5.2 (Residual representation split). *At the nonzero-coupling one-doublet target, F^{R_ϕ} has real dimension two and F_{mov} has real dimension two. The null space of K_ϕ lies in the fixed subspace. The response operator commutes with the residual action and preserves both subspaces.*

Proof. Use a unitary Higgs frame with ϕ in the second coordinate. The effective stabilizer consists of matrices $\operatorname{diag}(z, 1)$. Conjugation fixes the two diagonal infinitesimal directions and rotates the off-diagonal complex entry, a real two-plane. Because C_g is invertible, the transported representation on F has exactly the same fixed and nontrivial summands. The generator annihilating ϕ is diagonal, hence its pullback lies in the fixed subspace.

Equivariance of the representation gives $K_\phi R_\phi(t) = \rho(t)K_\phi$. Both actions preserve the respective metrics. Taking adjoints gives commutation of $2K_\phi^* K_\phi$ with $R_\phi(t)$. Averaging the latter action shows that M_ϕ commutes with the fixed-space projector. It therefore preserves its image and orthogonal complement. These conclusions are invariant under the original change of frame. $\square$

Definition 5.3 (Quadratic mechanism roles). *The independently specified mechanism requirements at this target are:*

1. *the full gauge-annihilator mechanism, consisting of all field directions whose action on the fixation section vanishes;*
2. *the full moving residual-action mechanism, consisting of all nontrivial residual-charge isotypic directions;*
3. *the neutral responsive mechanism, completing the first two gauge mechanisms orthogonally in the same kinetic space;*
4. *the scalar normal mechanism, orthogonal to the gauge orbit in the same Higgs fibre.*

Candidates are actual real subspaces of the indicated gauge or scalar space, not names, chosen bases or one-dimensional vectors in a charged plane. Completeness in the first two requirements means the full annihilator and full isotypic support. It is determined by the action, not by a desired number of occupants. The scalar and gauge spaces are separately typed even when eigenvalues happen to coincide.

Theorem 5.4 (Canonical electroweak–Higgs module decomposition). *There is exactly one simultaneous assignment of real subspaces to the four requirements in Definition 5.3. Its values are*

$$E_\gamma = \ker K_\phi, \quad E_W = F_{\text{mov}}, \quad E_Z = (E_\gamma \oplus E_W)^{\perp \kappa}, \quad E_H = T_\phi(G\phi)^\perp = \mathbb{R}\phi.$$

The gauge space decomposes orthogonally as $F = E_\gamma \oplus E_W \oplus E_Z$, with dimensions $(1, 2, 1)$. The scalar normal space has dimension one. The response form vanishes on E_γ and is strictly positive on $E_W \oplus E_Z$. Every target-preserving presentation comparison transports this entire decomposition, including its projectors and incidence maps.

Proof. The first requirement forces $E_\gamma = \ker K_\phi$; the second forces $E_W = F_{\text{mov}}$. By Lemma 5.2, the first is a one-dimensional subspace of the two-dimensional fixed space, and the second is the orthogonal two-dimensional moving space. The remaining neutral orthogonal complement is therefore one-dimensional. It is fixed by the residual action and contains no nonzero vector of $\ker K_\phi$. The positive-semidefinite identity $\kappa(a, M_\phi a) = 2\|K_\phi a\|^2$ proves positivity on the orthogonal complement of the kernel. The orbit/normal calculation forces the fourth value. These subspaces provide a simultaneous assignment, and the preceding forced equalities prove uniqueness without a count premise.

For covariance, let $T : F \rightarrow F'$ and $U : V \rightarrow V'$ be an actual comparison carrying the two metrics, coupling map and fixation section. Then $K_{\phi'}T = UK_\phi$, and the residual representations are intertwined. Thus T transports the kernel and fixed subspace. Preservation of the metrics transports their complements and projectors. The map U transports the orbit tangent and radial normal. This proves the claimed simultaneous covariance, not merely agreement of dimensions. $\square$

Corollary 5.5 (Global subbundles without a global unitary gauge). *If the licensed realization supplies a nowhere-zero Higgs section and compatible transition maps over its declared base, the local subspaces above glue to canonical real subbundles. No global Higgs frame is needed.*

Proof. On each overlap, the existing transition comparison satisfies the covariance conclusion of Theorem 5.4. Its subspace restrictions obey the original cocycle law. They therefore glue, and constant ranks follow from the nonzero-section and nonzero-coupling hypotheses. The result asserts no existence of such a section outside its stated realization domain. $\square$

5.2 Exhaustion of the joint-invariant supports

The natural candidate class consists of every real subspace of F invariant under the residual action and M_ϕ , independently of the mechanism names. These are precisely the possible supports of a quadratic gauge submodule closed under both operations. The scalar part is represented by the real quotient $N_\phi = V_{\mathbb{R}}/T_\phi(G\phi)$, identified with E_H by the retained metric. The combined candidate supports are graded: a gauge-field support and a scalar support are separately typed. This grading records their distinct tensor types, not a numerical mass distinction.

Theorem 5.6 (Exhaustive quadratic support classification). *Every real gauge subspace invariant under R_ϕ and M_ϕ is the direct sum of a subset of E_γ, E_W, E_Z . Consequently the lattice of graded invariant supports in $F \oplus N_\phi$ is the Boolean lattice on four minimal nonzero supports, the last being the scalar normal quotient. These supports are determined before particle names or carrier counts are assigned.*

Proof. Let $S \subseteq F$ be invariant under the two operations. Averaging the residual circle action gives a projector P_0 onto F^{R_ϕ} . It preserves S because every integrand does and S is finite-dimensional and closed. The spectral projector P_γ onto $\ker M_\phi$ also preserves S : the operator is self-adjoint, and this projector is the polynomial

$$P_\gamma = \prod_{\lambda \in \text{Spec}(M_\phi) \setminus \{0\}} \left(I - \frac{M_\phi}{\lambda} \right),$$

where each distinct nonzero eigenvalue occurs once. This construction does not assume that the charged and neutral nonzero eigenvalues differ. Commutation of the response and residual action gives commuting projectors, with $P_0 P_\gamma = P_\gamma$. Hence

$$P_W = I - P_0, \quad P_Z = P_0 - P_\gamma$$

are the orthogonal projectors onto the other two supports, and each preserves S . Their sum with P_γ is the identity. Thus S is the direct sum of its three intersections with the canonical supports. The neutral supports have dimension one, so their intersections are either zero or the entire line. The charged support is the real rotation representation of the connected effective residual circle. It has no proper nonzero invariant real subspace: the orbit of any nonzero vector spans the plane. Its intersection with S is therefore also either zero or the entire plane. Every subset sum is invariant, which proves both directions of the gauge classification.

The scalar quotient is one-dimensional, so it has just the zero and full subspaces. Grading prevents a graph between gauge and scalar directions from being a support of one tensor type. Taking the product of the gauge support lattice with these two possibilities gives the stated four-atom Boolean lattice. In particular, equality of a scalar and a vector quadratic eigenvalue does not merge their typed supports. $\square$

The classification is exhaustive within the quadratic module. Every higher-arity construction at this level uses sums of its four minimal supports or their typed relations; there is no fifth minimal support of the specified kind. Interacting composite operators, full source-response arguments and physical states are different objects. Tensor-product operators and additional scalar representations therefore require their own extension of the candidate class.

5.3 Role occupation at the module target

Assign the four mechanism roles to the minimal nonzero supports of Theorem 5.6, retaining the gauge and scalar sorts. Eligibility requires the corresponding mechanism property. Joint compatibility requires one common action, fixation section, pair of metrics, set of attachments and direct-sum relations. In particular, it excludes an assignment assembled from different parameter points. The union of compatible assignment images is the live residue of the Role-Occupancy Closure construction [13, Defs. 3.9–3.17].

Corollary 5.7 (Module occupation rigidity). *At each fixed common realization instance, the live module residue has exactly the four typed members E_γ, E_W, E_Z, E_H . There is exactly one compatible role-covering assignment, and its quotient by the full lawful target-preserving action is a singleton.*

Proof. The theorem supplies an eligible jointly compatible injective assignment: the three gauge summands are distinct nonzero orthogonal subspaces, and the scalar summand has a different sort. Every compatible assignment must have the same four values by the uniqueness proof. Their live union is consequently precisely this image, making the assignment bijective. All lawful automorphisms preserve the geometric requirements and hence the four subspaces. The singleton assignment therefore has one orbit. $\square$

The four occupants classify response modules, rather than fundamental carriers or physical particles. Complexifying E_W separates its two nontrivial residual characters, conventionally denoted W^+ and W^- . Complex conjugation relates these characters within the same real charged plane, without adding premetric gauge roots. The null module identifies the residual electromagnetic field direction; an asymptotic photon requires the separate state-space and continuation construction. The radial module

likewise precedes the positive-Hessian and scalar-pole questions, each of which has its own additional hypotheses.

Different admitted arguments or relative attachments in BCRS-I may induce the same four canonical modules. The result classifies the geometric image of the premetric atlas, not the fibers of that map. The complete-signature quotient remains the identity rule at each richer activated target [14, Thms. 4.4, 9.1 and 9.4].

6 Polarization, photon realization, and resonance authority

The null electromagnetic, charged and neutral vector, and radial scalar responses admit several distinct levels of physical description. Polarization geometry determines their kinematic degrees of freedom; state-space and continuation data determine whether they have an asymptotic particle or resonance realization. The underlying distinction connects SM-2's charge-response carrier, R7's complete localized persistence class, and SYM-2's separation of stable Wigner sectors from resonances [10, Sections 16–18] [11, Theorem 6.1] [12, Chapter 11, SYM2-T14]. The following constructions make each transition explicit while preserving its domain.

6.1 Polarization spaces as quotient responses

Work in the licensed, oriented and time-oriented Minkowski realization, with signature $(+ - - -)$ and units $c = \hbar = 1$. This is a downstream realization assumption; it does not replace the premetric carrier target. For a real momentum p , extend the Lorentz pairing complex linearly and write

$$p^\perp = \{\epsilon \in \mathbb{C}^4 : \eta(p, \epsilon) = 0\}.$$

For nonzero future null p and future timelike p , respectively, set

$$\mathcal{P}_0(p) = p^\perp / \mathbb{C}p, \quad \mathcal{P}_m(p) = p^\perp \quad (p^2 = m^2 > 0). \quad (6.1)$$

In the first space the Hermitian form is $h([\epsilon], [\zeta]) = -\eta(\bar{\epsilon}, \zeta)$; in the second it is the same expression without quotient brackets. The scalar response has the trivial complex line as its polarization fiber. These positive Hermitian spaces belong to real momentum orbits; complex resonance momenta require a separate analytic treatment.

Theorem 6.1 (Complete kinematic polarization fibers). *The spaces in (6.1) are positive Hermitian spaces of complex dimensions two and three. Proper orthochronous Lorentz transformations induce isometries between the corresponding fibers. For null p , the map*

$$[\epsilon] \mapsto p \wedge \epsilon \quad (6.2)$$

identifies $\mathcal{P}_0(p)$ with the field-strength image $\{p \wedge \epsilon : \epsilon \in p^\perp\}$ and removes exactly the gauge line. The null little group acts through its rotation quotient, with helicities $+1$ and -1 after complexification. The timelike little group acts as the three-dimensional vector representation of $SO(3)$.

Proof. Choose a Lorentz frame in which $p = (\omega, 0, 0, \omega)$ with $\omega > 0$. Orthogonality imposes $\epsilon^0 = \epsilon^3$. Subtracting $(\epsilon^0/\omega)p$ leaves the unique representative $(0, \epsilon^1, \epsilon^2, 0)$, and its Hermitian norm is $|\epsilon^1|^2 + |\epsilon^2|^2$. The radical of the unreduced form is therefore precisely $\mathbb{C}p$. Also $p \wedge \epsilon = 0$ holds exactly when $\epsilon \in \mathbb{C}p$, proving the field-strength claim. Null rotations change a transverse representative only by a multiple of p after passing to the transverse quotient, whereas ordinary rotations act on the transverse plane. The two complex circular directions have weights $e^{i\theta}$ and $e^{-i\theta}$.

For timelike momentum take $p = (m, 0, 0, 0)$. Orthogonality sets $\epsilon^0 = 0$, leaving the standard positive Hermitian form on $\mathbb{C}^3$, with the spatial rotation action. Lorentz transformations preserve $p^\perp$, send $\mathbb{C}p$ to $\mathbb{C}\Lambda p$, and preserve the displayed forms and wedge map. They therefore transport the constructions independently of the chosen frames. $\square$

The fiberwise construction defines a bundle on each momentum orbit. Local polarization frames are related by its transition maps, without requiring a global trivialization. The comparison data retain those maps, the little-group action, source attachments and domains. Comparing fibers at different momenta requires a specified Lorentz or continuation arrow, rather than a preferred identification inferred from their equal dimensions.

The polarization packet therefore comprises the momentum orbit, fiber metric, little-group action, local frame transitions, field-strength or response map, and every admitted continuation with its domain. It also records failure at zero momentum, a changed causal type or mass shell, a missing continuation, a singular presentation, or loss of the positive physical quotient. Within its domain the two null transverse classes and the three timelike spatial classes exhaust the vector polarizations. The massive longitudinal mode is included among the latter three. The familiar tally $2 + 3 \cdot 3 + 1$ thus checks the consistency of the response and polarization descriptions; it neither counts premetric carriers nor establishes physical poles.

6.2 The free helicity realization

Let $\mathcal{N}_+ = \{p : p^2 = 0, p^0 > 0\}$ and equip it with the Lorentz-invariant measure $d\mu(p) = d^3\mathbf{p}/(2|\mathbf{p}|)$. The bundle just constructed defines

$$\mathcal{H}_{\text{hel}} = L^2(\mathcal{N}_+, \mathcal{P}_0, d\mu). \quad (6.3)$$

For a proper orthochronous Poincare transformation define

$$(U(a, \Lambda)\psi)(p) = e^{ia \cdot p} \Lambda_* \psi(\Lambda^{-1}p). \quad (6.4)$$

Proposition 6.2 (Free massless helicity model). *Equations (6.3)–(6.4) give a positive-energy, strongly continuous unitary representation. It is the sum of the helicity +1 and −1 representations. Its symmetric Fock realization has stable free photon modes. This construction does not identify an interacting electroweak subspace with those modes.*

Proof. Lorentz invariance of $d\mu$ and the fiber isometry in Theorem 6.1 prove unitarity. The semidirect product law follows directly from Lorentz covariance of $a \cdot p$. The joint translation spectral measure is carried by $\mathcal{N}_+$; its spectrum is the closure of that cone. Energy is nonnegative, and the one-particle zero-momentum eigenspace is trivial. Strong continuity first holds for continuous sections of compact support in local bundle charts and then for all square-integrable sections by density and unitarity. The two circular eigenspaces of the oriented transverse rotation action are invariant line subbundles. The trivial action of null translations and the weights ± 1 identify their induced representations with the two helicity representations, using the Wigner classification in its stable-sector scope [12, Chapter 11 and proposition P-WIGNER-POINCARÉ]. Second quantization supplies the free Fock model; its Hamiltonian preserves particle number and has no decay interaction. None of these constructions supplies an identification with a coupled electroweak operator or scattering channel. $\square$

The construction realizes both photon helicities on a free branch. Its dynamics differ from those of the interacting electroweak target. An application to that target consequently requires an independent identification of its electromagnetic sector with this helicity model.

6.3 Source attachment and photon asymptotics

A photon interpretation of the residual response requires a map from the source-owned field-strength channel to a physical state space. Uniqueness of the charge direction and polarization fibers does not supply this map. The spectral and Gram data below determine it by completion, with asymptotic continuation retained as a separate input.

Definition 6.3 (Electromagnetic spectral and asymptotic attachment). *At the same fixed target, the attachment data consist of:*

1. *a positive physical Hilbert space with invariant vacuum Ω , strongly continuous positive-energy Poincare action, and the source-owned electromagnetic field-strength operator-valued distribution F on an invariant common domain containing Ω ;*
2. *a Poincare-invariant massless spectral projection E_0 orthogonal to the vacuum, together with the closed subspace $\mathcal{H}_\gamma = \overline{\text{span}}\{E_0 F(f)\Omega\}$;*
3. *a linear map I from the admitted test-function space to $\mathcal{H}_{\text{hel}}$, with dense range, such that*

$$\langle E_0 F(f)\Omega, E_0 F(g)\Omega \rangle = \langle If, Ig \rangle_{\text{hel}}, \quad (6.5)$$

and both maps transform covariantly under the same test-function action;

4. *independently established incoming and outgoing asymptotic maps on the photon scattering domain used by the target, preserving the inner product, Poincare action, source attachment, and declared infrared and boundary conditions.*

The massless spectral subspace and identity (6.5) are dynamical inputs. They are not inferred from the null response direction. The asymptotic maps are additional data, not consequences of a one-particle spectral projection.

Theorem 6.4 (Source-attached photon projection). *The data of Definition 6.3 determine a unique unitary map $V : \mathcal{H}_{\text{hel}} \rightarrow \mathcal{H}_\gamma$ satisfying $VI f = E_0 F(f)\Omega$. It intertwines the Poincare actions and induces the orthogonal photon projection onto $\mathcal{H}_\gamma$ in the physical Hilbert space. Together with the separately supplied asymptotic maps, it gives a stable massless photon/helicity projection on their declared domain. Lawful joint transport of all attachment data preserves it.*

Proof. Equation (6.5) makes the rule $If \mapsto E_0 F(f)\Omega$ well defined and isometric on a dense subspace: a zero-norm input has a zero-norm image. By completion it extends uniquely to an isometry, whose range is the closed span defining $\mathcal{H}_\gamma$ and is therefore all of that space. Covariance on test functions proves intertwining on the dense range of I ; boundedness extends it to the completed spaces. Composing the inclusion of $\mathcal{H}_\gamma$ with V and its adjoint gives the orthogonal projection. The target's asymptotic maps provide the separate scattering interpretation. Joint transport preserves the Gram identity, covariance, closure and asymptotic domains, so uniqueness of the extension makes the construction commute with that transport. $\square$

Identifying the neutral massless field-strength sector does not require a complete infrared construction of charged states. The photon asymptotic domain actually used nevertheless requires its own justification. Detector records are subsequent measurement projections, not definitions of E_0 or proofs of (6.5). SYM-8 organizes an *admitted* transverse photon pole, and SYM-2 classifies the corresponding stable sector. Their use here retains the independent target instantiation of the attachment data [8, Definition 2.2 and Sections 9.1–9.7] [12, SYM2-T14].

6.4 Gauge-independent continuation of simple poles

An unstable vector or scalar channel is represented by a pole of an authorized physical amplitude or invariant correlator, together with its sheet, residue and continuation data. This is a different object from a stable Wigner particle. A Nielsen-type identity controls how an existing pole is transported; its analytic consequence can be proved independently of the existence question.

Theorem 6.5 (Simple-pole transport on a fixed continuation sheet). *Let s range over a fixed connected complex neighborhood $\mathcal{O}$ on an authorized continuation sheet, and let ξ range over a connected real interval. Suppose the finite matrix $D(s, \xi)$ is holomorphic in s , continuously differentiable in ξ on the joint domain, and obeys*

$$\partial_\xi D = AD + DB, \quad (6.6)$$

where A and B are holomorphic in s and jointly continuous, with no coefficient singularity throughout the comparison domain. Write $d = \det D$ and assume $d(\cdot, \xi_0)$ is not identically zero for one ξ_0 . Then the zero set of d and each zero's multiplicity are independent of ξ .

Suppose, independently, that $d(s_*, \xi_0) = 0$ is a simple zero. Let J_L and J_R be analytic physical-channel maps, and assume

$$J_L(s_*, \xi_0) \operatorname{adj} D(s_*, \xi_0) J_R(s_*, \xi_0) \neq 0. \quad (6.7)$$

Then $J_L D^{-1} J_R$ has a genuine simple pole at s_* . If the channel maps are transported with the inverse-matrix factorization as specified below, the physical pole term, including its residue, is invariant. The theorem makes no pole-existence claim when the independent simple-zero or noncancellation premise is absent.

Proof. The adjugate identity, valid also at singular D , gives

$$\partial_\xi d = \operatorname{tr}(\operatorname{adj} D(AD + DB)) = (\operatorname{tr} A + \operatorname{tr} B)d.$$

Thus

$$d(s, \xi) = d(s, \xi_0) \exp \left(\int_{\xi_0}^{\xi} (\operatorname{tr} A + \operatorname{tr} B)(s, t) dt \right).$$

The exponential is holomorphic and nonzero on the common domain, so zeros and multiplicities are unchanged. In particular a pre-existing simple zero remains at the same point of the same sheet.

At ξ_0 , the adjugate formula and simplicity give residue

$$\operatorname{Res}_{s=s_*}(J_L D^{-1} J_R) = \frac{J_L(s_*) \operatorname{adj} D(s_*) J_R(s_*)}{\partial_s d(s_*)}, \quad (6.8)$$

which is nonzero by (6.7). Analytic background terms cannot cancel it.

To state the joint transport explicitly, solve $\partial_\xi L = AL$ and $\partial_\xi R = RB$ with identity initial values. Their finite-domain solutions are invertible, holomorphic in s , and yield $D(s, \xi) = L(s, \xi) D(s, \xi_0) R(s, \xi)$. If $J_L(s, \xi) = J_L(s, \xi_0) R(s, \xi)$ and $J_R(s, \xi) = L(s, \xi) J_R(s, \xi_0)$, then $J_L(s, \xi) D(s, \xi)^{-1} J_R(s, \xi)$ equals its value at ξ_0 away from the zeros. Equality of its meromorphic continuation gives equality of the residue and pole term. This final assertion requires the channel transport; it does not follow merely from invariance of the determinant zeros. $\square$

Regularity of the coefficients is essential; invertibility of A or B is not. Singular Nielsen coefficients, changes of analytic sheet, thresholds entering the neighborhood, and failure of physical channel transport are outside the comparison domain. The 1×1 case covers a scalar channel, while the matrix case allows neutral mixing. Repeated zeros retain their multiplicity, although the simple-pole residue formula does not apply to them. These distinctions specify the analytic content of the admitted pole covariance used in SYM-8 [8, Proposition 9.2].

6.5 Physical channels and realization domains

The invariant expressions $H^\dagger H$ and $H^\dagger i \overleftrightarrow{D}_\mu H$ identify candidate neutral scalar and vector channels. Realizing either as a physical pole requires the renormalized operator, state, analytic continuation and nonzero residue. A charged W channel instead requires an authorized charged-sector construction, dressing, or physical scattering amplitude with the corresponding source and boundary attachment; a neutral operator cannot provide it. This distinction between invariant channels and elementary field coordinates is also central to SYM-8 [8, Sections 9.4 and 9.7–9.10].

For W^+ , W^- , Z and the radial Higgs channel, the resonance data retain one common target and source map, the operator or physical amplitude, the state and renormalization domain, and the continuation sheet and neighborhood. They also retain an existing pole with its order and nonzero physical residue, its gauge and presentation transport, and the relevant stability, infrared and boundary qualifications. Only with these data does a pole on the selected unphysical sheet with negative imaginary part have the stated unstable-resonance interpretation. The quadratic response eigenvalue alone does not determine that object.

SYM-8's bundle $\mathcal{S}_{\text{pert}}$ contains these spectral premises as target certificates. Its reconstruction theorem fixes their organization once they are instantiated at the target. BRST cohomology removes unphysical coordinate occupants, while isolated pole existence remains a separate dynamical requirement [8, Definition 2.2 and Theorem 9.3].

Proposition 6.6 (Separated realization status). *On the licensed Minkowski branch of the common electroweak response target, the polarization fibers and free helicity realization above are constructed. An interacting photon projection is obtained on the domain of Definition 6.3; an unstable-pole projection is obtained only on the domain of the complete physical-channel packet. Where those data have not been established, the corresponding response remains structural and the physical projection is unactivated. This is not a theorem that the photon or pole fails to exist.*

Proof. The positive constructions follow from Theorem 6.1 and Proposition 6.2. The photon attachment is Theorem 6.4. The pole packet supplies the independent existence, channel and continuation premises, while Theorem 6.5 establishes the stated presentation invariance in its analytic subdomain. Without those premises the maps have no licensed value. Their partiality neither alters the upstream response construction nor implies physical nonexistence. $\square$

The resulting distinction is between the constructed polarization and free-helicity realizations and their additional interacting spectral or asymptotic attachments. BCRS-I, SM-2, R7, SYM-2 and the conditional SYM-8 theorem do not by themselves instantiate those attachments on the common target. The polarization and transport results remain valid on their stated domains, while the physical projections retain the independent existence requirements above. Field labels, detector records, numerical masses and representative propagators do not replace those requirements.

7 Interactions, process types and physical continuation

Interactions couple the canonical response supports. Their local coefficients follow from the admitted kinetic and interaction terms; a scattering or decay interpretation additionally requires continuation between physical states or channels with the appropriate boundaries. This distinction permits the interaction structure to be established before its physical process domains are complete.

7.1 Local vertices forced by the doublet kinetic term

In a local radial chart $H = (0, v + h)^T / \sqrt{2}$, the nonderivative gauge part of the Higgs kinetic term is

$$\frac{1}{2} \left(1 + \frac{h}{v} \right)^2 \mu_\phi(A, A).$$

The fixation, current normalization and couplings are the same as in the quadratic response. The radial chart gives a convenient calculation of the invariant multilinear coefficients.

Proposition 7.1 (Radial–vector interaction descendants). *At $v \neq 0$, the kinetic term determines the radial–vector–vector and two-radial–two-vector coefficients by polarization of μ_ϕ/v and $\mu_\phi/(2v^2)$, respectively, when “coefficient” means the coefficient of the monomials hA^2 and h^2A^2 . The actual multilinear derivatives are $D_h D_a D_b \mathcal{L} = 2\mu_\phi(a, b)/v$ and $D_h^2 D_a D_b \mathcal{L} = 2\mu_\phi(a, b)/v^2$ at zero. Each coefficient vanishes when a gauge argument lies in E_γ , and is nonzero on the appropriate paired charged or neutral responsive directions. The coefficients are covariant under simultaneous transport of the fixation, kinetic and current data.*

Proof. Expand $\frac{1}{2}(1 + h/v)^2 \mu_\phi(A, A)$ into terms of radial degree zero, one and two. The coefficients are the displayed quadratic forms. The response-kernel theorem gives $\mu_\phi(a, b) = 0$ for every b when $a \in E_\gamma$; this follows either from the Gram identity or from self-adjointness. Positivity on the responsive complement gives nonzero pairings there. The expression is built from the common invariant kinetic term, so transporting all its inputs transports its multilinear derivatives. A change of scalar normalization must also transport h and v ; it is not a change of just one coefficient. $\square$

The vanishing result is specific to the local coefficients of the renormalizable kinetic term. A physical $h \rightarrow \gamma\gamma$ amplitude may involve an induced operator or a loop construction with its own internal arguments, and is not covered by that result. A local hVV coefficient likewise identifies an interaction without establishing the kinematic or spectral availability of an on-shell decay.

Proposition 7.2 (Residual charge grading of interactions). *Use an actual retained residual circle subgroup or its specified lift acting on all of the admitted source representations, not merely the effective Higgs action. Complexify these representations and decompose them into its character spaces. An invariant multilinear vertex can have a nonzero restriction to a tuple of character spaces only when the total character is trivial. Conjugate arguments carry the contragredient character. This rule is preserved by contraction of paired internal legs.*

Proof. For infinitesimal residual charges $q_1, \dots, q_n$, equivariance multiplies the scalar vertex value by $e^{it \sum_j q_j}$ along the actual one-parameter action. These charges need not be integers in electron-charge units. They are integral weights only after the true period of the retained circle action has been normalized. Invariance for every t implies that a nonzero value has $\sum_j q_j = 0$. The same statement can be expressed without a weight normalization by requiring the product of the characters to be trivial. Evaluation against a dual internal leg cancels the two opposite characters, so contraction preserves the rule for the remaining external legs. $\square$

Charge neutrality is necessary for a nonzero vertex. The inherited representation, Lorentz tensor type, operator order and source coefficients impose further restrictions. Charged-current incidences retain their distinct matter-current arguments: an interaction between two carriers does not identify them.

7.2 A source-evaluated process family

The complete local operator grammar supplies source vertices as invariant multilinear derivatives of the admitted kinetic, potential and interaction functionals. Their fields, derivative orders, supports and boundary restrictions are inherited from those functionals, rather than specified by a desired amplitude. The preceding propositions construct nonzero vertices and establish their charge restrictions.

Definition 7.3 (Typed process diagrams). *A process profile records a common realization instance, ordered input and output field or state sorts, a process kind, and its boundary and continuation data. The kinds distinguish propagation, charged-current or neutral-current interaction, production, decay, and record formation. A diagram consists of finitely many actual source vertices and admitted internal continuation maps with these external sorts. Evaluation is defined only when all internal maps exist on their stated domains and the contractions share the same source and boundary data.*

An algebraic vertex diagram uses field-response sorts. A physical process diagram also has the state, resonance or inclusive-channel authority needed for every physical endpoint and every continuation. A record diagram includes its measurement or reconstruction map. These are different types even if their familiar endpoint labels coincide.

Theorem 7.4 (Exact process composition and covariance). *The source-evaluated diagrams compose on their exact common-boundary and continuation domain. Composition is associative whenever both parenthesizations are admitted by the same source composition laws. Lawful comparisons transport domains and evaluations. The resulting family is indexed by the full realization and authority data; it does not acquire physical endpoints by forgetting that index.*

Proof. Composition joins matching external legs by the given continuation map, retaining the original vertices, boundary restrictions and source instance. The evaluation is the corresponding composite multilinear operation. Associativity follows from the actual composition and contraction laws on the common domain. No assertion is made for singular, divergent or otherwise unlicensed contractions. Every primitive comparison intertwines its vertex or continuation map and preserves its domain. Induction on the finite diagram therefore proves both domain preservation and the commuting evaluation equation for the full composite. The induction carries the external sort and authority index unchanged. It supplies no map from a field sort to a physical state sort not already present among the primitives. $\square$

An invertible comparison of realization profiles transports every vertex, endpoint and boundary, and hence reindexes the process diagrams. Identities and composites inherit the source comparison laws. This is the bosonic counterpart of the typed process family in FCGS-III [2, Sec. 7], with physical processes restricted to their actually established domains; the unresolved full physical apex of that paper remains unresolved here.

Corollary 7.5 (Processes do not multiply carrier identities). *A vertex, propagating edge, decay relation or reconstruction map is not an additional carrier or an independent physical pole solely by being a process component. A physical process claim cannot be obtained from the algebraic diagram by dropping a missing authority argument.*

Proof. The domain and codomain of a process component are already typed source objects. Its structural role is a relation between those objects, not a source construction of a new identity class. Promotion would require the independent identity and realization maps of the claimed new object. The evaluation induction retains all primitive domains, so an absent physical endpoint or continuation does not become available by composition. $\square$

The kinetic and representation structure thus determines a nonempty local interaction family. Its scattering, decay, resonance and detector interpretations depend on the corresponding physical

continuation and authority. The local vertices remain defined when one of those interpretations is unavailable, without thereby establishing the missing physical realization.

8 Terminal comparison and deletion-minimal response support

The complete quadratic module classification supports a precise minimality theorem, but terminal physical identity requires a richer comparison. Its observations include every admitted physical continuation, source and boundary restriction, channel identity and failure value. A common carrier or module rank does not determine these observations. Complete signatures connect the two levels and provide the criterion for lossless deletion at either one. The following sections construct the required native observation algebra and give explicit whole-response reconstructions. The criterion below therefore serves both the quadratic support result and the later functional classification.

8.1 Complete signatures on each activated domain

For an activated realization target T , let $\mathcal{X}_T$ be its independently source-admitted presentations, and retain the complete target observation family and licensed partial continuations. These include definedness: an undefined continuation, a defined failure, and a successful value are different outcomes. A complete contextual signature $\Sigma_T(x)$ records every such outcome after every admitted finite context. Mixed contexts retain all their actual sibling arguments and attachments.

Proposition 8.1 (Terminal comparison without an upstream singleton). *On a fully formed, target-complete observation domain, the unique standing-correct terminal identity quotient, where standing-correct means both witness soundness and observational completeness, is*

$$\mathcal{Q}_T = \mathcal{X}_T / \sim_T, \quad x \sim_T y \iff \Sigma_T(x) = \Sigma_T(y).$$

Every licensed operation descends on its exact domain. For a live subfamily A_T , its quotient cardinality is $|\Sigma_T(A_T)|$. An unformed physical observation domain does not have a terminal quotient whose cardinality can be reported as zero or one.

Proof. Equality of the complete signatures preserves the domain and value of each operation followed by every continuation, since appending that operation is itself an admitted context. It therefore gives congruence with domain reflection. Any standing-correct equivalence must identify signature-equal inputs by completeness and cannot identify signature-distinct inputs by soundness. This gives the unique quotient and its bijection with the signature image. These are the identity principles of SCIC [14, Thms. 4.4, 9.1 and 9.4] with the inherited contextual partial-domain discipline of BCRS-I. If the observation or continuation domain is not formed, the displayed definition has no such instance; a missing object is not an empty quotient. $\square$

Pointwise eligibility, simultaneous compatibility and the lawful assignment action are fixed on the same target. Role occupation uses the union of compatible assignment images and then the compatible role-covering assignment quotient [13, Defs. 3.9–3.17]. The quadratic support theorem computes this quotient as a singleton. A richer physical target adds observations that must enter its own evaluation. Even an established photon projection or resonance packet does not imply equality of the complete signatures of every interacting terminal presentation.

8.2 Lossless deletion and semantic reconstruction

Theorem 8.2 (Semantic deletion criterion). *Let $d : \mathcal{X}_T \rightarrow \mathcal{Y}$ be a specified well-typed erasure at a fixed activated target. There exists a map $s : d(\mathcal{X}_T) \rightarrow \text{im } \Sigma_T$ satisfying $\Sigma_T = s \circ d$ if and only if*

$$d(x) = d(y) \implies \Sigma_T(x) = \Sigma_T(y) \quad \text{for every } x, y \in \mathcal{X}_T.$$

When it exists, s is unique. Thus a pair with equal retained data and different standing observations proves nonrecoverability even under arbitrary reconstruction from those retained data.

Proof. Factorization immediately gives the implication. Conversely, assign to a point $d(x)$ the value $\Sigma_T(x)$. The implication makes this independent of every choice of preimage, and every point of the stated image has a preimage. This defines the unique map with the required equation. Definedness and failure coordinates are included in the signature, so the conclusion does not allow a reconstruction that preserves values while changing their domains. $\square$

Lossless deletion depends on the information in the retained data, not on preservation of a particular representation. For example, deleting an explicit copy of μ_ϕ while retaining the full action, coupling map and Higgs metric is lossless: the Gram construction recovers it. A chosen neutral basis can be omitted when the response, kinetic and current maps still determine the invariant decomposition. The moving-space projector can likewise be recovered by averaging the retained residual action. Such reconstructions use the same target data and introduce no repair.

Erasing an actual input, attachment or physical continuation may instead destroy information. Nonrecoverability requires the equal-retained-data pair, or an independent nonfactorization argument, specified in the theorem. Absence of evidence for a pole is a different matter and does not establish physical impossibility.

8.3 Deletion-minimality of quadratic amplitude support

Consider the licensed Minkowski quadratic branch with positive radial Hessian. Fix the source, all parameter values and kinetic forms, and actual nonzero momenta on the indicated positive-energy shells. Using the polarization spaces already constructed, define the graded complex amplitude space

$$\begin{aligned} \mathcal{A}_x = & (E_{\gamma, \mathbb{C}} \otimes \mathcal{P}_0(p_\gamma)) \oplus (E_{W, \mathbb{C}} \otimes \mathcal{P}_{m_W}(p_W)) \\ & \oplus (E_{Z, \mathbb{C}} \otimes \mathcal{P}_{m_Z}(p_Z)) \oplus (E_{H, \mathbb{C}} \otimes \mathbb{C}). \end{aligned}$$

The scalar momentum is on its own positive-mass shell, even though its one-dimensional polarization space is written simply as $\mathbb{C}$. The masses here are the positive quadratic coefficients at this branch, not asserted interacting pole masses. The tensor-product metrics and orthogonal sum give a positive Hermitian amplitude norm. This norm is an admitted invariant observation of the quadratic response space. The construction is at fixed momenta; its momentum-dependent version is the corresponding bundle, not a claim that a single sharp-momentum vector is a normalizable scattering state.

Theorem 8.3 (No-repair minimality of the isotypic amplitude support). *Each of the four displayed summands is nonzero. Erasing any one entire summand prevents reconstruction of the full amplitude norm and therefore of any complete quadratic-response signature that contains that norm. The amplitude support is deletion-minimal at this whole-isotypic-summand granularity, even when all source parameters and reconstructible projectors are retained.*

Proof. The internal support dimensions are $(1, 2, 1; 1)$, and the respective polarization dimensions are $(2, 3, 3; 1)$. Each tensor product is therefore nonzero and positive. Choose a nonzero vector a entirely

in the summand being erased. It has the same retained image as the zero vector, with every source and parameter datum unchanged. Yet its total squared norm is strictly positive, while that of zero is zero. The nonfactorization direction of Theorem 8.2 excludes reconstruction. The argument applies separately to each summand. The source determines the available subspace, but does not determine the independently variable amplitude inside it; reconstructing a projector cannot recover the erased amplitude. $\square$

The charged real isotypic support contains two conjugate complex charge components. They are retained together at the stated granularity; resolving the charges permits a finer minimality test. The result establishes the indispensability of actual quadratic response amplitudes, rather than independence of the source premises or of every field in an interacting terminal packet. Physical spectral and asymptotic extensions have their own complete signatures and corresponding deletion tests.

8.4 Extensions and scope preservation

Zero fixation, vanishing coupling, a nonpositive radial Hessian, a different scalar representation, higher-dimensional operators, a changed global form and new boundary sectors define explicit branches or target changes. The one-doublet result determines which parts of its decomposition survive comparison with them; it does not exclude these physical possibilities. An extra scalar field can introduce a realization degree of freedom without constituting a second primitive Higgs identity. Its source and complete signature decide that distinction.

On every branch, the predecessor carrier and mechanism identities remain intact. A downstream physical measurement may constrain an already licensed realization, but cannot replace a missing source attachment, complete terminal signature or physical continuation.

9 The native electroweak observation and continuation algebra

The quadratic decomposition is a consequence of a larger mathematical object: the source-attached electroweak interaction law and its admitted responses. Its continuation algebra contains complete local vertices, current pairings, source restrictions and boundary maps. On an analytic or state-space realization it also contains the actual channel functions or state-continuation operators. The distinction matters even at fixed particle labels: equal masses do not imply equal interaction vertices, and equal pole positions do not imply equal amplitudes.

We now construct this algebra on the exact received source domain. The construction starts from native operations, not from a desired number of occupants. It makes explicit both the data needed to reconstruct a whole response object and the observations that are allowed to distinguish its standing identity.

9.1 Native objects and source reception

Fix an activated realization target T , its actual source, operator order, global form, boundary conditions, spectator records and branch. Parameter values may vary over the source-admitted image unless T fixes them. A physical branch is present only where its independent physical data are supplied. In particular, omission of a spectral realization is not a spectral realization with zero spectrum.

Definition 9.1 (Native response interfaces). *The native EW/H interfaces are the following whole-object interfaces, with their actual incident maps.*

1. *The local interaction interface consists of the source-attached gauge and scalar differential laws at the declared operator order, kinetic and current pairings, selected fixation branch, and all inherited*

source, anomaly, vacuum and deformation restrictions. It includes boundary terms or vacuum-value data whenever the target admits an observation of them.

2. An admitted analytic interface consists of source-labelled physical channel functions with their argument spaces, current insertions, connected analytic sheets, continuation maps, singularity and boundary domains, and the actual comparisons identifying different lawful presentations of those channels.
3. An admitted state-continuation interface consists of the labelled physical state construction, its scalar products, continuation operators and their domains, and any supplied incoming, outgoing or asymptotic comparison maps.

The source, current, restriction and boundary arrows joining these interfaces are part of the data. Derivatives, response matrices, polarization spaces and particle-channel readouts are operations or reducts of these interfaces; they do not become additional independent whole-object roles merely because they can be displayed separately.

These interfaces are specified by their mathematical work, independently of a preferred representative. The local interface is the complete operator, current and fixation structure received by SYM-8. Its analytic and state-space statements use the separate physical bundle of that source [8, Defs. 2.1–2.2, Secs. 9–10 and Def. 13.1]. The broader source-first symmetry theorem tests candidate membership against these precise source domains and retains the entire response/spectrum object, not only its rank or named particle entries [17, Secs. 2.1–2.3].

Definition 9.2 (Primitive native operations). *For the interfaces actually supplied at T , the primitive operations are:*

1. the inherited BCRS-I source-witness, restriction, transport, loop, representation, charge, anomaly, vacuum, deformation, and joint Higgs–electroweak operations;
2. evaluation, variation, differentiation, tensor contraction and composition of the declared local interaction laws, with the admitted quotient and field-presentation maps;
3. evaluation of the supplied physical channels on their complete labelled inputs, their actual analytic continuations and boundary values, and the admitted extraction of poles, residues or channel comparisons;
4. scalar-product, operator and continuation evaluations in the supplied state interface, including actual domain tests and any incoming–outgoing comparison;
5. the actual incident source, current, boundary and restriction maps, and the target’s compatibility, obstruction and report operations.

Every operation retains its exact domain and all structural parameters. Its arguments range over the native admitted families, not a finite sample. A local or physical quantity used only in a gauge-fixed calculation is read through its admitted quotient or physical comparison; its raw coordinate is not thereby an extra terminal observation.

Proposition 9.3 (Source reception and native-operation coverage). *On the exact source-received EW/H domain, every native finite response construction is a finite diagram of the operations in Definition 9.2, with its actual common interfaces. This includes restriction, partial response, continuation, mixed incidence and failure constructions. The family is not restricted by the availability of a named normal-form representative.*

Proof. An independently supplied native realization first restricts to its actual upstream source diagram. The common-source construction of Section 2.3 receives that diagram without reconstructing source

witnesses from a bare quotient. BCRS-I's atomic coverage and source-generation theorem cover every lawful witnessed instance of its source operations, including their actual mixed restrictions [1, Secs. 3, 16–17].

For the realization part, the declared local operator grammar fixes which field, connection, kinetic, potential and current objects may be used. Its local responses are evaluations and variations of those laws; their compositions, restrictions and quotient maps are precisely the second family above. SYM-8's physical interface separately retains the actual amplitudes, correlators, boundary charges and state or pole authority used in a physical assertion. Their native evaluations and continuations therefore belong to the third and fourth families, with their original domains. The full phase interface includes the admitted invariant correlation and response object, boundary conditions and regime, so this reduction does not replace that object by a finite particle list [8, Secs. 8–10 and Def. 13.1].

The source-first symmetry transfer treats the target-qualified candidate family before consulting named atlas rows. It tests every exact source premise, preserves the action, Higgs orbit, stabilizer, full response and spectrum, quotient, anomaly, vacuum, operator and regime coordinates, and assigns each source-normalized solution its own lawful class. Thus a source-satisfying candidate absent from a published list still enters the same construction. A failed mandatory equation, changed target coordinate or unproved source premise retains its distinct disposition; none is silently identified with an admitted value [17, Sec. 2.1 and electroweak bridge in Sec. 2.2].

Finally, inspect a finite native response diagram one vertex at a time. Each vertex is an operation on one of these source or realization interfaces, an incident map, or a compatibility/failure test. Replacing it by the corresponding received operation leaves all actual edges and arguments in place. Induction on the finite diagram proves the claim. This is coverage of the exact received source target, not an assertion that every mathematical model sharing its particle notation satisfies the source hypotheses. $\square$

9.2 Reconstruction of the complete local law

The local operation family has an exact finite description when its degree and differential order are fixed. The data to be reconstructed are the entire laws, not merely their quadratic terms at the vacuum.

Theorem 9.4 (Polynomial-jet reconstruction). *Let J and F_0 be finite-dimensional real vector spaces and let $P : J \rightarrow F_0$ be a polynomial of degree at most d . Fix a marked jet $z_0 \in J$. Then the finite tensor family*

$$\mathcal{T}_d(P, z_0) = (D^k P(z_0))_{0 \leq k \leq d}$$

determines the entire law by

$$P(z_0 + \eta) = \sum_{k=0}^d \frac{1}{k!} D^k P(z_0)[\eta, \dots, \eta].$$

Consequently it determines every finite evaluation, variation, differentiation, contraction or composition of that law. For a differential-polynomial bundle law of finite jet order, the same conclusion holds when the coefficient tensors are retained as sections over the full declared base with their transition maps.

Proof. Choose linear coordinates centred at z_0 . Each component of P is a sum of monomials of degree at most d . Differentiating a monomial at the origin recovers its coefficient with the corresponding factorial, and all higher derivatives vanish. Summing the recovered monomials gives the displayed finite Taylor identity. Hence two such families agree exactly when the polynomial laws agree. Evaluation and all the listed finite operations can then be performed on the recovered law, proving the first conclusion.

For a bundle law, apply the argument in each local jet chart at each base point. Retaining the coefficient sections permits their base derivatives to be taken wherever the given regularity allows. On

an overlap, the two recovered expressions are the original law transformed by the actual jet-coordinate change, so they agree under that change and glue. Gauge changes of connection jets use the prolonged gauge transformation, including its derivatives; they are not replaced by fibrewise conjugation of bare partial derivatives. The chain rule therefore transports the reconstructed operations correctly. $\square$

Corollary 9.5 (Complete local-law presentation fibre). *At a fixed differential-polynomial interaction law and its actual bundle, source and boundary maps, complete coefficient presentations form a nonempty connected groupoid under their lawful chart comparisons. Their evaluations reconstruct the same entire law and every native local response, not only the gauge-action Gram form.*

Proof. The coefficient tensors of the given law provide a presentation. Every other complete presentation reconstructs that same law by Theorem 9.4. The identity of the underlying law, expressed through the two retained chart systems, gives the lawful comparison; it preserves the attached bundle, source and boundary maps. The resulting comparison is defined on all local charts and their overlaps, hence connects the presentations. All evaluated responses agree by reconstruction. $\square$

The finite degree bound is essential. Even all derivatives at one point do not determine an arbitrary smooth function: the function $\exp(-1/x^2)$, extended by zero at $x = 0$, has the same jet there as the zero function. Variable coefficient sections must likewise be retained on their base, not replaced by a finite sample. Analytic channel reconstruction instead uses convergent germs and their actual continuation domains; no convergence of a merely formal series is inferred from the polynomial theorem.

9.3 Joint incidence and fixed-sibling comparisons

Definition 9.6 (Intrinsic operation graph). *For a primitive operation f , its intrinsic graph comprises its typed argument spaces, exact domain D_f , output map, and all incident source restrictions and structural parameters, after only the already lawful source and presentation identifications. An absent domain, an undefined argument, a defined zero, a defined obstruction and a successful nonzero value remain distinct. A comparison of graphs must preserve and reflect D_f , not merely agree where both maps happen to be defined.*

Theorem 9.7 (Relative operation-graph descent). *A source-authorized redescription of one input port preserves the domain and output of every incident native operation when the other actual inputs and the intrinsic mixed structural parameter are fixed and the incident maps are the actual transports certified by the native comparison or common intrinsic descent. The result extends to every finite mixed context. An isomorphism of marginal port objects alone is not such a redescription. Comparing two independently supplied joint diagrams additionally requires their actual joint incidence fidelity; it is not licensed by redefining one diagram's arrows.*

Proof. First take an invertible source-authorized port redescription $u : X' \rightarrow X$. Its certified incident transports are $m' = mu$ for an outgoing map, $a' = u^{-1}a$ for an incoming map, and $A' = u^{-1}Au$ for an internal endomorphism. These equalities describe the actual transported diagram; they are not definitions imposed on an independently supplied second diagram. Every incident square therefore commutes, while no map or object belonging only to another sibling is changed. The native comparison identifies domain membership by pullback through u , and its inverse proves reflection.

For source quotient or compression maps, BCRS-I supplies a common intrinsic descended diagram and exact domain pullback, including the fixed-input mixed Higgs–gauge operation on its complete intrinsic pointed ports. Use that descended diagram, not an inverse on raw source representatives [1, Sec. 16]. Local-law comparisons preserve evaluations by Theorem 9.4; physical comparisons retain their

actual channel, state and continuation authority. Boundary and current maps are transported by the same incident-square rule.

At a composite operation, apply this result at the first vertex meeting the changed port and then at its output. Each step preserves and reflects its domain and transports the output needed by the next step. Induction gives the result for every finite context, at any input position, including contexts with repeated or shared source restrictions. For independently supplied diagrams, a marginal isomorphism does not establish any of the incident equalities above. Those equalities must instead follow from their actual joint-probe comparison or an already proved whole-interface reconstruction. $\square$

Proposition 9.8 (Exact common-diagram assembly). *The native source and realization graphs assemble through their actual common interfaces. Their incidence completion introduces only images of admitted core data along admitted paths, and their joint assembly identifies only the relations generated by the supplied overlaps. It creates neither a physical source at an unsupported sort nor a comparison between unrelated source charts. The assembled object is an incidence ledger; it does not assert that an arbitrary colimit of physical algebras is a new physical algebra.*

Proof. Represent an incidence image by a triple consisting of a core object, an admitted path and a core element. Postcomposition gives its further incidence maps. Identify triples only by the relations already supplied by the core incidence diagrams. On a common interface take the joint image of the actual overlap maps. The tagged coproduct of the completed cores, divided by the least equivalence generated by these overlap images, has the required assembly map. Any compatible cocone factors through it by sending a tagged element to its cocone image; agreement on the overlap relations makes this well defined, and every element has such a representative, proving uniqueness.

An unsupported sort has no representing core element and path. An additional identification is absent from the least generated relation. Thus neither can enter this construction. This is the explicit source-colimit argument of Module D, specialized to the received EW/H incidence schema, with no import of that module's role count [18, Thms. 5.7, 5.17–5.18]. The primitive operations remain the already received operations on their actual common domains. Their functionality and physical admissibility are not inferred from cocompleteness of a set-valued diagram category. $\square$

9.4 The reconstructed algebra and terminal identity

The primitive readouts are the target-authorized output maps of Definition 9.2: intrinsic source witnesses, invariant local responses and current pairings, complete admitted channel and state evaluations, boundary and continuation outputs, and their domain, compatibility and failure values. Auxiliary frames, coefficient displays, proof certificates and source filenames are not additional readouts. The values of a many-input readout retain the actual attachment and all its labelled arguments.

Theorem 9.9 (Native reconstruction and factorization). *On the exact received target, retain the intrinsic source operation graphs, the reconstructed whole local law, every supplied physical channel and state-continuation object, and their actual joint incidence diagram. Then:*

1. *every native finite use is reconstructible from these objects, on its exact domain;*
2. *lawful comparisons of the reconstructed objects preserve all native observations and every mixed continuation;*
3. *the equality relation determined by all primitive readouts after the reconstructed native continuations is precisely the full native observational equivalence.*

Thus the terminal observation algebra is constructed from explicit native operation families; no unspecified complete-signature object is required as an additional input.

Proof. By Proposition 9.3, a native finite use is a finite diagram of the listed operations. At an upstream vertex, BCRS-I reconstructs the source operation from its full intrinsic law graph. At a local vertex, the polynomial-jet theorem reconstructs the whole differential law; in the normalized one-doublet specialization, the invariant local parameter reconstruction gives a smaller equivalent description. At a physical vertex, the actual channel or state-continuation object supplies the full function or operator and its domain. Its analytic or Gram normal form may replace that object only through the corresponding whole-object reconstruction theorem. At an incident vertex, the retained source, current or boundary map is used. Composition of these evaluations proves the first assertion.

Theorem 9.7 proves comparison covariance and domain reflection at each vertex, including any one-input slice with the siblings fixed. The same induction therefore proves the second assertion. It also shows how to form the continuation algebra explicitly by composing the reconstructed maps, rather than presuming that a complete family of final answers was supplied.

Let two candidates agree on every primitive readout after every such continuation. Any native observation is an output of the finite diagram just reconstructed, hence one of those readouts after its preceding continuation, and has the same status and value on the two candidates. Conversely, each retained readout followed by an admitted continuation is itself a native use. A difference in one such value or domain status is therefore an actual separating native observation. These implications give exactly the full observational equivalence. They do not add raw coefficient equality as a stronger identity test. $\square$

Corollary 9.10 (Activated native standing quotient). *The independently received native operation family and its reconstructed continuations activate the terminal standing interface on their exact domain. Its quotient is the image of the native readout evaluation map, with all domain and failure values retained. Every signature-distinct pair is separated by an explicit primitive readout after an admitted continuation. Any sufficient reconstruction presentation may be further quotiented only when that deletion preserves these values.*

Proof. The source coverage theorem exhausts the received operation types, and the factorization theorem exhausts their native finite uses. The joint incidence construction retains all actual restrictions and overlap relations. Hence definedness, invariant consequences, compatibility, failure and continuation are represented before any carrier count is formed. SCIC's activation and identity-closure theorems apply to this constructed interface [14, Thms. 3.4 and 4.4]. The quotient is the evaluation-image quotient, and unequal evaluations identify the separating coordinate. A finer reconstruction presentation has no authority to retain an invisible auxiliary distinction. $\square$

This construction separates two questions. Complete local coefficient data or full channel germs can reconstruct an entire interface while still containing a removable representation choice. Terminal identity retains only its admitted consequences. Conversely, once an admitted consequence differs, its class cannot be collapsed by common carrier provenance. The source-first symmetry construction makes the same distinction by evaluating the physical projection independently before matching source-conclusion branches [17, Sec. 2.5].

The result is uniform over the actual admitted parameter image. Positive couplings and a stable polynomial minimum construct an open local-action family, but do not by themselves supply every source, boundary or spectral antecedent at every parameter value. Admissible deformations must preserve their operational witnesses, quotient, bounded envelope, shared source and vacuum data [16, Def. 3.6 and Thm. 6.4]. The exact full-native image therefore remains the one formed from actual admitted source instances. Its complete terminal comparison is now available; its cardinality and role occupation are subsequent evaluated questions, not consequences of the number of displayed modules.

10 Reconstruction of the local interaction response

The four canonical supports do not exhaust the information in an electroweak interaction. Two realizations can have exactly those supports and nevertheless have different current strengths or scalar vertices. This section determines that additional information on the canonically normalized one-doublet local branch. Its coordinates are response observations in a fixed comparison, not a numerical selection of the parameters of Nature.

Definition 10.1 (Normalized local comparison family). *Fix the metric and differential comparison, actual gauge-group global form, doublet and current representations, canonical gauge and scalar kinetic normalizations, and the standard normalization $y_H = 1/2$, $Q = T_3 + Y$ used for the canonical response formulas. Retain the actual spectator, Yukawa, boundary and topological data as fixed sibling inputs. These are their independent input tensors, not derived masses or other outputs which vary with v . In particular, theta terms and their boundary observations are not erased. The variable local gauge–Higgs action is the standard one-doublet action with positive couplings g_1, g_2 , $v > 0$, and scalar polynomial*

$$V(H) = a + bH^\dagger H + c(H^\dagger H)^2, \quad c > 0, \quad b = -cv^2.$$

The sign conventions are fixed by the retained current attachment. Allowed comparisons are source-preserving changes of presentation in this normalized family. The local observation target includes its quadratic forms, normalized current coupling, and all local action coefficients, equivalently their finite field-jet derivatives. Such coefficients are tensor-valued observations with their source-attached field and current frames transported; their raw entries in unrelated frames are not compared. Equivalently, the calculations may be made in one retained common source frame. A value-sensitive target also includes the scalar value at fixation; a derivative-only target compares the scalar polynomial modulo an additive constant. These are different targets.

The displayed family is a local-action family. Its intersection with the admitted native source image is denoted by $\mathcal{L}_T$. It is the stated minimally coupled variable action, not the class of all covariant operators of the same engineering dimension. An additional scalar–curvature term which changes the stationary equation belongs to an enlarged operator target. Numerical response coordinates are expressed in the retained units throughout. No inference from a positive coefficient tuple to the existence of every required global, boundary or spectral source witness is made. Conversely, the classification below applies to every native admitted point on this local branch without another response-equality premise. The distinction between invariant response data and a raw numerical coupling coordinate is the one established in the gauge-coupling analysis [15, Thms. 3.9, 5.1 and 5.5–5.6; App. A10–A11].

10.1 An explicit inverse from response observations

At a point of the family define

$$w = \frac{g_2^2 v^2}{4}, \quad z = \frac{(g_1^2 + g_2^2)v^2}{4}, \quad q = \frac{g_1^2 g_2^2}{g_1^2 + g_2^2}, \quad h = 2cv^2, \quad \ell = a - \frac{cv^4}{4}. \quad (10.1)$$

Here w, z, h are the charged, neutral and radial quadratic response eigenvalues in the retained kinetic normalization. They are not asserted to equal interacting physical pole locations. The quantity q is the squared coefficient of the normalized residual electromagnetic current along the unit kinetic null line. Thus its definition includes the current attachment; it is not extracted from the null eigenvalue alone. Squaring removes the lawful simultaneous sign choice of that line and its attached field, not an independently changed current. Finally, ℓ is the retained scalar value at the stationary fixation. Whether it is an observation depends on the declared target, not on the absence of a local scattering vertex.

Theorem 10.2 (Constructive local interaction normal form). *For the value-sensitive normalized family, the response map in (10.1) is a bijection from the positive coefficient family to*

$$\Omega_5 = \{(w, z, q, h, \ell) : 0 < w < z, q > 0, h > 0, \ell \in \mathbb{R}\}.$$

Its inverse is

$$g_2^2 = \frac{qz}{z-w}, \quad g_1^2 = \frac{qz}{w}, \quad v^2 = \frac{4w(z-w)}{qz}, \quad (10.2)$$

$$c = \frac{h}{2v^2}, \quad b = -cv^2, \quad a = \ell + \frac{cv^4}{4}. \quad (10.3)$$

The positive square roots recover the conventionally normalized g_1, g_2, v . For the derivative-only scalar target the same statement holds with Ω_4 , obtained by omitting ℓ , and with the scalar polynomial taken modulo its additive constant.

Proof. Positivity gives $w > 0$, $z - w = g_1^2 v^2 / 4 > 0$, $q > 0$ and $h > 0$. Dividing the formulas for w and $z - w$ by the formula for z , and substituting into the formula for q , gives the two expressions for the squared couplings. The equation for w then gives the stated v^2 . The radial Hessian fixes $c = h/(2v^2)$, stationarity fixes $b = -cv^2$, and the scalar value fixes a .

Conversely, for any point of Ω_5 , all three squared quantities in (10.2) are strictly positive, and the recovered c is positive. Inserting the inverse into (10.1) returns the same five coordinates. The recovered potential has its radial minimum at $H^\dagger H = v^2/2$ and is bounded below. Both compositions are therefore identities. If the constant is unobserved by the declared scalar target, the same calculation determines precisely its equivalence class modulo constants, proving the four-coordinate statement. $\square$

Theorem 10.3 (Complete local-response classification). *Two points of the value-sensitive normalized local family have the same complete local gauge–Higgs response observations if and only if their five response coordinates agree. With the fixed sibling data retained, every such local observation factors uniquely through Ω_5 . On the actual admitted source image the terminal local quotient is exactly*

$$\Omega_T = \{(w, z, q, h, \ell)(x) : x \in \mathcal{L}_T\} \subseteq \Omega_5,$$

not a stipulated singleton. The corresponding statements hold for the derivative-only target with four coordinates.

Proof. Equality of the five observations reconstructs the same normalized couplings, fixation and scalar polynomial by Theorem 10.2. The gauge representation, kinetic forms, current conventions, spectator and topological inputs are the same retained siblings. The complete local action and its local gauge–Higgs coefficients are consequently the same. All finite field-jet derivatives, contractions and admitted algebraic compositions of those coefficients agree, including domains and failure values of any prescribed partial algebraic operation. Conversely, each of w, z, q, h, ℓ is an admitted observation at this target. Equal complete local observations therefore force equality of the five coordinates. The inverse constructs the factorization on the full family, and restricting it to the actual image gives its unique factorization on Ω_T . The same argument omits only the scalar constant at the derivative-only target. $\square$

This is more than a classification of a mass matrix. The inverse fixes the scalar self-interactions and the normalized gauge–Higgs and gauge-current vertices, as well as their quadratic coefficients. It is less than a classification of arbitrary interacting physical responses without their analytic and state-space data. A higher operator, a different theta or boundary term, a new Yukawa input, or a new physical continuation changes the target or adds a retained coordinate. It cannot be hidden in the phrase “same five masses and couplings.”

10.2 No-repair minimality on the family target

Theorem 10.4 (Irredundant local response packet). *On the family-level target Ω_5 , none of the five response coordinates can be deleted while preserving all local response observations, even if reconstruction from the other four is arbitrary. No coordinate is duplicated in an otherwise retained pointed coefficient certificate. The same assertion holds for the four coordinates of Ω_4 . On a restricted admitted image Ω_T , deletion of a specified coordinate is lossless exactly when that coordinate is a function of the retained ones on Ω_T .*

Proof. Fix a point $(w, z, q, h, \ell) \in \Omega_5$. The following single-coordinate replacements remain in Ω_5 :

$$w \mapsto w/2, \quad z \mapsto z + w, \quad q \mapsto 2q, \quad h \mapsto 2h, \quad \ell \mapsto \ell + 1.$$

For each deletion separately, use the corresponding pair of points. The retained four coordinates agree and the deleted admitted observation differs. The explicit inverse constructs two members of the normalized positive local-action family for each pair. The semantic deletion criterion (Theorem 8.2) excludes every reconstruction from the retained data. The first four witnesses give the result on Ω_4 .

On an arbitrary admitted image, the forward implication of the criterion requires constancy of the deleted coordinate on each retained-coordinate fibre, which is exactly the stated function condition. Conversely, that function recovers the deleted coordinate; the complete inverse then recovers every local observation. No unproved assertion that the displayed mutation pairs belong to every native source image is used. $\square$

The comparison target in this theorem allows its response coordinates to vary. If an independently retained pointed certificate already contains their exact values, deleting a second copy is lossless. This is not a counterexample to family-level minimality; it is a different erasure with more information retained. Similarly, if a source theorem restricts the admissible image to a proper relation among these invariants, the image-relative criterion, rather than the open-family witness, decides which coordinates remain essential. This is the precise distinction between a unique assignment within each response fibre and uniqueness of the physical response itself.

Corollary 10.5 (Local chart and broader target). *For every activated broader EW/H target whose normalized local restriction is the one above, the map to Ω_T is a well-defined local observation of its terminal class. Any broader identification which equates two different points of Ω_T is not observation-preserving. Equality in Ω_T alone does not identify broader classes with different retained source, boundary, spectral or continuation data.*

Proof. The broader observation family contains the indicated local probes, so broader equality implies equality of their values. The converse requires all its other observations as well; the local reconstruction does not erase them. $\square$

11 Reconstruction of physical response and continuation data

The physical response of a fixed carrier need not be a finite list of pole positions. An energy-resolving observation family can retain a whole spectral function without adding a primitive carrier. This distinction is explicit in the spectral-witness analysis of [19, Proposition 3.7 and Remark 3.8]. We now construct determining invariants for the analytic channels and physical state realizations admitted by the common target. The construction retains the joint source attachment required by [8, Definitions 3.1–3.2 and Section 9] and supplies a concrete instance of target-relative interface completeness [14, Definition 3.3 and Theorem 3.4].

11.1 Full analytic germs and physical channel reconstruction

Let X be a connected complex manifold carrying the target's specified analytic sheet, and let $p \in X$ be a marked regular point with a fixed local coordinate $z = (z_1, \dots, z_d)$. Fix the finite-dimensional source and current ports V_α, W_α , including their normalization. The admitted physical channels are actual meromorphic maps

$$F_\alpha : X \longrightarrow \text{Hom}(V_\alpha, W_\alpha)$$

that are holomorphic near p . Finite-dimensional multilinear channel maps are included by using their tensor-product input ports. The singular loci, authorized continuation paths, boundary procedures, source labels and definedness or failure conditions retain their source-owned meanings. In particular, analytic continuability does not by itself authorize a physical continuation.

For a multi-index $\nu \in \mathbb{N}^d$, define the coefficient tensor

$$A_{\alpha, \nu} = \frac{1}{\nu!} \partial_z^\nu F_\alpha(p). \quad (11.1)$$

The germ includes every coefficient, not only the principal parts at selected poles. On a disconnected analytic domain, a marked germ is retained on each connected component. The sheet and path attachment is part of the domain; different branches are not identified through an unmarked local formula.

Theorem 11.1 (Complete analytic channel reconstruction). *On the common marked analytic domain just specified, equality of all coefficient tensors in (11.1) is equivalent to equality of the actual meromorphic channel maps. Together with equal source, continuation, boundary and failure data, it preserves every admitted operation on those maps. In particular it preserves evaluation, differentiation, finite algebraic composition, fixed-port contraction and specified boundary limits. On an admitted one-variable continuation it preserves the full Laurent expansion, including the regular background, and hence every physical pole, order and residue in that continuation.*

Proof. In the chosen coordinate neighborhood, each entry of F_α equals its convergent multivariable Taylor series. Equal coefficient tensors therefore give equality on an open neighborhood of p . The identity theorem for meromorphic functions on the connected complex manifold X , applied entrywise, gives equality everywhere as meromorphic maps. One can see the continuation explicitly by covering any path by overlapping coordinate neighborhoods: equality propagates across each overlap, and local denominators reduce the meromorphic comparison to the holomorphic identity theorem. Equality of the maps conversely gives all the displayed coefficients.

Every admitted operation on equal maps has the same value. For boundary limits this statement uses the same limiting procedure and the same target topology, with its authority and failure conditions retained. A one-variable meromorphic restriction has a unique Laurent expansion on each admitted punctured neighborhood, so all of its coefficients agree. If ℓ, r are fixed sibling current inputs, then $\ell F_\alpha r$ agrees with $\ell F'_\alpha r$ without changing either input. For a multilinear channel the coefficient tensors act on the same full tensor-product port; keeping all their components likewise preserves every fixed-sibling evaluation. $\square$

Under a lawful coordinate, channel-basis or normalization change, the coefficient tensors, source ports and marked analytic domain are transported jointly before comparison. This retains the same physical function rather than comparing components in unrelated frames. A nonzero matrix residue may vanish after contraction with a particular current pair. Thus the relevant pole test is a pole of the attached physical channel, or a nonzero contraction of its full port tensor, as in Theorem 6.5.

The regular coefficients are essential on the unrestricted analytic family. The functions $(s - a)^{-1}$ and $(s - a)^{-1} + c$, with $c \neq 0$, have the same pole and residue but different responses. Nor does an

arbitrary finite jet suffice: adding $\epsilon(s - s_0)^{N+1}$ preserves the jet through order N while changing a later coefficient. These comparisons explain why a finite pole list cannot replace the full analytic sort.

11.2 A finite determining jet on the rational branch

Finite reconstruction is available when the source independently bounds the rational response degree. The bound is an algebraic property of the supplied response law, not a hypothesis that its finite jet is already complete.

Theorem 11.2 (Bounded rational-response reconstruction). *Let $F = P/Q$ and $F' = P'/Q'$ be scalar rational functions satisfying*

$$\deg P, \deg P' \leq N, \quad \deg Q, \deg Q' \leq M, \quad Q(s_0)Q'(s_0) \neq 0.$$

Equality of their Taylor coefficients at s_0 through order $N + M$ implies $F = F'$. Thus $N + M + 1$ coefficients determine the function within this bounded rational class. The result applies entrywise to finite matrix channels with the same bounds.

Proof. The difference $F - F'$ has a zero of order at least $N + M + 1$ at s_0 . Multiplication by QQ' , which is nonzero there, gives the same vanishing order for $PQ' - P'Q$. This polynomial has degree at most $N + M$ and hence is zero. The rational functions therefore agree. $\square$

Corollary 11.3 (Polynomial inverse-response bound). *Let $D(s)$ be an $n \times n$ polynomial matrix with entries of degree at most m and determinant not identically zero. Let the attached polynomial current maps J_L, J_R have degrees at most ℓ, r , and let the polynomial background B have degree at most k . Then the actual channel*

$$F(s) = J_L(s)D(s)^{-1}J_R(s) + B(s) \tag{11.2}$$

has a rational denominator of degree at most $M = nm$ and numerator degree at most

$$N = \max\{\ell + (n - 1)m + r, k + nm\}.$$

At a point where $\det D(s_0) \neq 0$, its jet through order $N + M$ determines the channel in this bounded class.

Proof. The identity $D^{-1} = \text{adj } D / \det D$ gives denominator degree at most nm and cofactor degree at most $(n - 1)m$. Combining the two terms of (11.2) over this denominator gives the asserted numerator bound. Cancellations only lower the degrees. Theorem 11.2 then applies to each entry. $\square$

This branch includes finite polynomial inverse-response constructions with polynomial currents and background. General interacting self-energies or branch cuts use the full analytic germ unless an independent reduction establishes the bounded rational hypothesis.

11.3 A joint kernel for states, currents and continuations

The physical state sort is specified independently of the invariants used to compare it. Fix its source and boundary tags, test-vector ports, bounded action and continuation generators, and admitted finite-word composition rules. In a realization, let v_λ be the actual vector indexed by a source tag and an admitted word. The family includes vectors from all current, field-strength, incoming and outgoing attachments in the same physical Hilbert space. Extra labels for operator graph pairs will be specified below. Exact domain and failure information accompanies every partial source operation.

The Hilbert space to be reconstructed is

$$\mathcal{H}_{\text{cyc}} = \overline{\text{span}\{v_\lambda : \lambda \in \Lambda\}}, \quad K(\lambda, \mu) = \langle v_\lambda, v_\mu \rangle. \quad (11.3)$$

One joint complex kernel contains every cross-tag entry. For a bounded generator T_a , the label action records $T_a v_\lambda = v_{a\lambda}$ on an admitted labelled span dense in its Hilbert source sort. A fixed external port map is recorded on its source-owned dense test family with its specified continuity, or through its full closed-graph data. Where a source map is given on every admitted test vector, those input labels are kept exactly. No independent change of the test-vector coordinates is implicit.

Proposition 11.4 (Construction from a positive source kernel). *Let K be a Hermitian positive-semidefinite kernel on the fixed source-word label set Λ . Require every specified linear source-word relation to have zero K -norm. For each bounded generator a , let Λ_a be its admitted input labels, with specified output label $a\lambda$, and suppose a finite nonnegative constant M_a satisfies*

$$\sum_{i,j} \bar{c}_i c_j K(a\lambda_i, a\lambda_j) \leq M_a^2 \sum_{i,j} \bar{c}_i c_j K(\lambda_i, \lambda_j) \quad (11.4)$$

for every finite family $\lambda_i \in \Lambda_a$. Require their labelled classes to have dense span in the completed Hilbert source sort. For an external Hilbert port $\mathcal{V}_t$, let D_t be its specified dense test subspace and let $\lambda(t, u)$ label the assigned image of $u \in D_t$. Suppose a finite nonnegative constant C_t satisfies, for every finite family $u_i \in D_t$,

$$\sum_{i,j} \bar{c}_i c_j K(\lambda(t, u_i), \lambda(t, u_j)) \leq C_t^2 \left\| \sum_i c_i u_i \right\|_{\mathcal{V}_t}^2. \quad (11.5)$$

Then K constructs a cyclic Hilbert realization of the labelled vectors, with bounded generators of norm at most M_a and bounded external port maps of norm at most C_t . The supplied linear relations hold in this realization. Its labelled Hilbert realization is unique up to the unitary in Theorem 11.5.

Proof. Let $\mathcal{F}_\Lambda$ be the complex vector space of finitely supported formal sums of symbols e_λ . Define

$$\left\langle \sum_i c_i e_{\lambda_i}, \sum_j d_j e_{\mu_j} \right\rangle_K = \sum_{i,j} \bar{c}_i d_j K(\lambda_i, \mu_j).$$

Positivity gives the Cauchy–Schwarz inequality for this semidefinite form. Consequently its zero-norm space $\mathcal{N}$ is its radical and is a linear subspace. The quotient $\mathcal{F}_\Lambda/\mathcal{N}$ is a pre-Hilbert space. Complete it to $\mathcal{H}_K$ and set $v_\lambda = [e_\lambda]$. These vectors have Gram kernel K and dense linear span. The required source-word relations vanish because their formal sums lie in $\mathcal{N}$.

On the admitted formal input span, send e_λ to $e_{a\lambda}$. Inequality (11.4) sends every zero-norm input to a zero-norm output, so the map descends through the null quotient, independently of the chosen formal representative. The same inequality bounds its norm by M_a . Density therefore gives its unique bounded extension to the Hilbert source sort.

For the external port, send a finite sum $\sum_i c_i u_i$ to $\sum_i c_i v_{\lambda(t, u_i)}$. If the input sum is zero, (11.5) makes the output norm zero. All linear relations among the external inputs are thus preserved, and the assignment defines a linear map on D_t . The inequality bounds its norm by C_t , giving a unique extension to $\mathcal{V}_t$. The joint-kernel comparison theorem proves the stated uniqueness of the completed labelled realization. $\square$

An actual source realization with bounded generators and port maps supplies these inequalities by its operator-norm bounds. Conversely an abstract positive kernel satisfying them gives the stated mathematical realization, not automatic admission as an electroweak source. Physical source, covariance, spectral, boundary and continuation requirements remain the target’s independent conditions; none is created by positivity alone.

Theorem 11.5 (Simultaneous joint-kernel reconstruction). *Two realizations on this same labelled source grammar have equal kernels in (11.3) if and only if there is a unique unitary $U : \mathcal{H}_{\text{cyc}} \rightarrow \mathcal{H}'_{\text{cyc}}$ such that $Uv_\lambda = v'_\lambda$ for every label. This unitary simultaneously intertwines all supplied bounded generators. It also intertwines all bounded current, incoming and outgoing maps from the fixed external ports whose values are specified on dense labelled inputs. It preserves every admitted finite word, mixed matrix element and norm-continuous process limit. Preservation of their native partial-operation domains, boundary permissions and failure conditions additionally requires equality, or the specified joint transport, of that independent domain and status information. Equality of K alone does not impose those permissions.*

Proof. Define on finite labelled sums

$$U_0\left(\sum_i c_i v_{\lambda_i}\right) = \sum_i c_i v'_{\lambda_i}.$$

Equality of the kernels gives equality of the squared norms of every such sum and its proposed image. In particular every zero vector has zero image. The map is well defined and isometric, with dense domain and dense range in the two cyclic spaces. Its unique continuous extension is a unitary U .

On the labelled span, $UT_a v_\lambda = Uv_{a\lambda} = v'_{a\lambda} = T'_a Uv_\lambda$. Boundedness extends this equality to the completed Hilbert source sort. For a bounded external port map E_t , the labels give $UE_t u = E'_t u$ on a dense family of the *same* external inputs u , so $UE_t = E'_t$ everywhere. The single unitary therefore intertwines all these maps at once. Compositions, inner products and norm limits preserve the resulting equalities. The domain and failure data are compared as part of the same native object, so this step does not enlarge any authorized continuation. Conversely a labelled unitary preserves all kernel entries. Its values on a dense span also prove uniqueness. $\square$

Several Hilbert sorts are handled by their orthogonal direct sum, with sort tags retained. External Hilbert ports are anchored by their fixed dense input vectors, so the reconstructed unitary acts as the identity on those ports. The theorem identifies the entire physical Hilbert space only when the source proves that its labelled span is cyclic for that space. An invisible orthogonal summand is not reconstructed by an observation family that never reaches it. Strongly continuous covariance actions can be included among the bounded unitary generators; their simultaneous intertwining then preserves the supplied covariance and spectral representation.

11.4 Closed operators and their source domains

A dense algebraic word domain does not alone determine a closed operator. The self-adjoint generators in the physical covariance target retain their actual domains [12, Chapters 7–8]. Two concrete kinds of supplied data are sufficient for exact reconstruction.

Proposition 11.6 (Reconstruction of closed operators and maps). *Let U be the simultaneous unitary of Theorem 11.5, and let A, A' be the actual closed operators on the reconstructed cyclic Hilbert sorts being compared. A larger ambient space is included only when its cyclicity has been supplied independently. Either of the following source-supplied conditions gives*

$$U \text{Dom}(A) = \text{Dom}(A'), \quad UA = A'U \quad \text{on } \text{Dom}(A). \quad (11.6)$$

1. *A common $z \in \rho(A) \cap \rho(A')$ is supplied, and the bounded resolvents $R_z = (A - zI)^{-1}$ and $R'_z = (A' - zI)^{-1}$ are included among the intertwined generators.*
2. *The joint labelled family includes matched graph pairs (u_i, Au_i) and $(u'_i, A'u'_i)$ whose finite linear spans are dense in the respective full closed graphs, with the product Hilbert norm. Both components participate in the same joint kernel.*

The second conclusion also holds for closed maps between different Hilbert sorts, using the corresponding reconstructed unitaries on the source and target. For an anchored external source port, its unitary is the identity.

Proof. In the resolvent case, $UR_z = R'_zU$ gives

$$U \operatorname{Dom}(A) = U \operatorname{Ran} R_z = \operatorname{Ran} R'_z = \operatorname{Dom}(A').$$

Writing $x = R_z y$, we have $Ax = zx + y$ and $A'Ux = zUx + Uy = UAx$. This proves (11.6).

In the graph case, $U \oplus U$ sends each labelled pair (u_i, Au_i) to $(u'_i, A'u'_i)$. It therefore maps their finite linear spans onto one another. By continuity and the stipulated graph-density it maps the full closed graph of A onto that of A' . Equality of the graphs gives equality of the domains and operator values as in (11.6). For maps between two sorts, the same proof uses the source unitary direct sum the target unitary; an anchored source port is unchanged. $\square$

For a self-adjoint operator a nonreal resolvent parameter is available. The graph alternative does not presume that a general closed operator has a nonempty resolvent. A graph-dense family is a means of reconstructing the actual graph, not a new identity coordinate: different choices spanning the same closed graph yield the same operator. The choice of a core or a proof of its density is not retained as physical content. Additional authorized boundary restrictions are nevertheless retained; the reconstructed operator does not enlarge the source's allowed continuation domain.

The need for actual domain information can be seen on $L^2(0, 1)$. The self-adjoint operators

$$A_\theta = -i \frac{d}{dx}, \quad \operatorname{Dom}(A_\theta) = \{f \in H^1(0, 1) : f(1) = e^{i\theta} f(0)\}$$

agree on $C_c^\infty(0, 1)$ but have spectra $\theta + 2\pi\mathbb{Z}$ and different domains for distinct θ modulo 2π . Integration by parts gives the stated self-adjoint boundary condition. The graph-norm closure of the compactly supported smooth functions is $H_0^1(0, 1)$, not the full displayed domain. Thus their common algebraic action does not supply the graph-density premise.

11.5 Mixed attachment and observable kernels

The joint kernel is essential when incoming and outgoing data are compared. On $\mathbb{C}^2$, take the fixed external ports and

$$W_{\text{in}} = I, \quad W_{\text{out}} = \operatorname{diag}(1, e^{i\theta}).$$

Each map is unitary, so both marginal kernels are independent of θ . The cross kernel instead determines $S = W_{\text{out}}^* W_{\text{in}} = \operatorname{diag}(1, e^{-i\theta})$. At the fixed external vector $u = (1, 1)/\sqrt{2}$, the probability $|\langle u, Su \rangle|^2 = \cos^2(\theta/2)$ varies. Theorem 11.5 uses a single unitary for both maps and therefore retains this relative attachment.

For the same reason, covariance of a jointly transformed tuple does not by itself justify replacement of one input while its siblings are held fixed. Such a replacement must preserve either the full joint kernel including those siblings, or the reconstructed source interface anchored to their unchanged common ports. All actual closed sibling maps then use the same graph and domain under Proposition 11.6.

The passage from a reconstructing kernel to a terminal physical invariant also requires observational faithfulness. SYM-8 retains the *admitted* invariant correlation data in its phase signature, not unrestricted inner products of arbitrary gauge-fixed vectors [8, Equation (24) and Theorem 13.2]. The following criterion identifies a positive source branch on which kernel reconstruction uses those observations themselves.

Proposition 11.7 (Descent from admitted invariant correlations). *Suppose the target admits a positive-metric vacuum realization and an adjoint- and product-closed family of invariant source words A_λ on a common invariant domain containing Ω . Suppose its admitted correlation observations include $\langle \Omega, A_\lambda^* A_\mu \Omega \rangle$ for every pair of labels under comparison. Then the joint kernel of $v_\lambda = A_\lambda \Omega$ is recoverable from those admitted observations. Kernel equality is therefore equivalent to equality of this correlation family, and Theorem 11.5 classifies its cyclic realization up to the labelled unitary.*

For additional current, asymptotic or graph labels, the same conclusion holds precisely when their cross-kernel entries and domain data are likewise admitted or independently reconstructed from admitted source observations.

Proof. The common invariant domain makes the source products well defined, and the adjoint identity gives

$$K(\lambda, \mu) = \langle A_\lambda \Omega, A_\mu \Omega \rangle = \langle \Omega, A_\lambda^* A_\mu \Omega \rangle.$$

Every entry of the kernel is thus an admitted observation, and every observation in this stated family is that entry. Their equalities coincide, so the reconstruction theorem applies. Adding a label preserves the same conclusion when its additional entries and domain data satisfy the stated recovery condition; without that condition the extended kernel would contain more information than the given observation family. $\square$

The positive-metric invariant local-field branch is supported by the canonical interface in [12, Section 8.2]. Charged superselection sectors are separately typed there. Arbitrary coherent cross-sector probes, graph-domain tests or unrestricted source-word closure are consequently not added by notation. When the source admits fewer observations, the above theorem remains a reconstruction of the supplied whole interface; terminality for that smaller observation family requires its own descent argument.

Even on a cyclic positive representation, there is generally no universal finite word-length truncation. For example, let T be multiplication by $e^{i\phi}$, let $\Omega = 1$, and use

$$d\mu_\pm(\phi) = (1 \pm \epsilon \cos(L\phi)) \frac{d\phi}{2\pi}, \quad 0 < \epsilon < 1, \quad L = 2N + 1.$$

Both representations on $L^2(S^1, \mu_\pm)$ are cyclic for T, T^* . The joint Gram entries of words of length at most N agree because their Fourier differences have magnitude at most $2N$, whereas $\langle \Omega, T^L \Omega \rangle = \pm \epsilon/2$. This is a counterexample in the unrestricted cyclic family, not an assertion that both measures are realized by a particular electroweak source.

11.6 The reconstructed physical interface

Proposition 11.8 (Complete supplied physical interface). *Fix the actual analytic and cyclic physical source sorts described above, including their common current and state attachment, native operation domains, continuation paths and boundary conditions. Their full channel germs and single joint kernel, with the supplied resolvent or full-graph data where required, reconstruct the analytic functions and physical realization up to the one source-preserving unitary. They therefore preserve every native operation on these reconstructed objects, including fixed-sibling mixed comparisons at the anchored ports.*

If the analytic coefficient observations and the kernel/domain information descend to the admitted physical observation family, as in Proposition 11.7, and the native source constructors are covered by these sorts, this reconstruction is observation-complete on that source domain.

Proof. Theorem 11.1 identifies the actual analytic maps. Theorem 11.5 and Proposition 11.6 identify the states, generators, port maps and exact closed domains simultaneously. Common source attachment keeps the relation between analytic and state realizations fixed, rather than matching them independently.

Every native operation on equal analytic objects or unitarily identified physical objects has the same output, with partiality preserved by the domain information. Anchored ports keep the other inputs fixed in a mixed comparison. Finally, the stated observation descent makes the reconstruction data recoverable from the physical observations, while source coverage ensures that no admitted native object or operation is omitted. These two directions give observation completeness on the declared domain. $\square$

The two descriptions need not be independently irreducible. If a channel is certified to be a matrix element of the reconstructed resolvent and fixed current vectors, its germ is recoverable from the state/operator interface. Such a channel is a derived output, not an additional essential input. Conversely a source may supply an invariant analytic channel without supplying a Hilbert-space realization of every response. The reconstruction then applies to that analytic sort on its own domain.

These theorems classify actual supplied physical responses; they do not infer an interacting spectral state or an unstable pole from quadratic nullity or a response eigenvalue. The independent existence requirements in Section 6 remain unchanged. Once those objects are supplied, however, their complete functional and continuation content can be reconstructed without constructing a larger interacting theory than the target requires and without introducing another primitive carrier.

12 Whole-interface realization and role occupation

The preceding reconstructions concern entire operation-bearing objects: a local interaction law, an admitted family of analytic channels, and a source-generated state representation with its continuations. They permit a role-occupancy calculation for the full activated diagram, rather than only its four quadratic supports. The calculation is fibred over actual response data. Its uniqueness concerns realization of each such datum and does not identify different response data.

12.1 Roles fixed by the source work

Definition 12.1 (Activated whole-interface diagram). *Fix one native EW/H comparison target with its common source anchors, allowed operation and continuation types, actual field and current interfaces, and boundary and failure tests. Its whole-interface diagram retains the independently required operation-bearing objects and their generating incidences. The realization interface types are:*

1. *the covariant local interaction law, including the required field-jet operations and its actual source/current restrictions;*
2. *each jointly admitted analytic-channel family, with its channel labels, connected marked domains, sheet and continuation data;*
3. *each jointly admitted cyclic state-and-continuation interface, with its source-word labels, inner products and actual bounded actions, closed graphs or source resolvents.*

Only interfaces independently activated by the source target are included. The state or analytic role is not silently populated on a structural-only target. Repeated uses of the same interface share that object. A derivative, matrix entry, support projector or residue is an output of its operation-bearing object unless the source independently requires a separate interface for additional work. The inherited carrier laws and metric/global/boundary data are the source anchors, not newly postulated primitive occupants.

Write I_T for the resulting source-required role index. It is formed before evaluating candidates. It need not have a universal cardinality: several channel interfaces, or no activated physical one, are both permitted. Distinct indices specify distinct typed interface work, not new elementary particle species.

The generating incidences are the actual source and current restrictions, differentiation into vertices and quadratic responses, source-word and continuation actions, mixed Gram and channel contractions, state/channel realization maps, analytic path composition, closed-graph or resolvent restrictions, and boundary and status maps. All their compositions retain the exact native domains. This is the source-first distinction between generic interface work and fidelity to one source diagram; it is not a list of names selected to match a proposed number of occupants. Primitive-status exhaustion [10] does not replace these richer operation burdens.

For an actually admitted native instance u , let $b(u)$ be its intrinsic reconstructive packet: its local tensor or invariant coordinates, full admitted channel germs and continuation data, joint source-word Gram and closed-domain data, and the actual incidence data linking them to the native source anchors. Lawful chart choices are already identified; source filenames, proof names and choices of a graph core are not coordinates. Where one block is derivable from others, it can be retained for comparison without being claimed necessary. The actual reconstructive packet image is

$$\mathcal{B}_T = \{b(u) : u \text{ is admitted at the fixed target } T\}.$$

The parameter point is not chosen by a role count. The local restriction of this image is Ω_T of Theorem 10.3 when the normalized local branch applies. The broader image can retain genuine global, state and continuation distinctions not present in that local restriction.

An intrinsic reconstruction datum need not be an observation of a coarser physical target. We therefore do not identify $\mathcal{B}_T$ with the terminal quotient without the detectability proof below. In general its reconstructed readout kernel must still be taken. A mathematical superselection coherence or untested operator domain does not become a physical observation by being retained in a reconstruction.

Definition 12.2 (Candidate work, fidelity and compatibility). *A generic candidate for a role in I_T is a mathematical interface capable of performing that role's specified operations: respectively an admitted covariant local law, a meromorphic channel family on the required domains, or a cyclic representation with the required operator domains and labelled source actions. Its generic type is specified without a preferred reconstruction or a desired terminal equivalence class.*

At $b \in \mathcal{B}_T$, a candidate is eligible when its source attachments are lawful and its elementary reconstruction data have the values specified by the corresponding restriction of b . These data are tensor evaluations, full channel germs, joint Gram entries, domain/status outputs and the source-owned incidence probes; terminal-signature equality is not an eligibility premise. They are called observationally faithful only when their recovery from the target's admitted observations has been proved. Candidates are compared by actual source-preserving interface isomorphisms. Their disjoint typed union, quotiented by those comparisons, is S_b .

An eligible complete injection $a : I_T \rightarrow S_b$ is compatible when its generating incidences assemble on the given common source diagram and satisfy every retained mixed, boundary, continuation and failure equation. This condition is not replaced by rolewise eligibility. Lawful automorphisms preserve the whole source and probe tuple, including relative attachments.

The probe requirement has mathematical content: the reconstruction theorems lift equality of these elementary data to a comparison of the entire interface. If a target permits noncyclic hidden summands, undetermined self-adjoint extensions, unspecified analytic sheets or unobserved additional local operators, those theorems do not apply to that enlarged candidate family. The corresponding additional work must be received before its role can be declared exhausted.

12.2 Reconstruction of mixed incidences

Lemma 12.3 (Joint reconstruction preserves the generating diagram). *On the received local, analytic and cyclic domains, two eligible whole-interface presentations with the same packet b admit lawful*

comparisons commuting with all generating incidences. The comparisons preserve and reflect their exact domains and status values. This holds chartwise on the actual source atlas; no universal amalgamation of unrelated native charts is assumed.

Proof. For a normalized local law, the inverse in Theorem 10.2 reconstructs the same source-attached action and all its derivatives. For a more general received polynomial law use its complete tensor reconstruction on the declared base. A lawful change of frame transports the entire law and its jet prolongation, so equality is not raw equality in unrelated charts. Local source restrictions and differentiated vertices therefore commute with this comparison.

For each admitted channel family, Theorem 11.1 recovers the same functions on the retained marked domains and sheets. Its source and channel labels, path data and failure domains are retained. Thus contractions by the given currents and allowed analytic continuations agree, including continuation failure rather than only successful common-domain values.

For a cyclic interface, Theorem 11.5 gives the unique unitary carrying every labelled generated vector to its counterpart. It is one joint unitary, not a collection of independently chosen marginal ones. Source actions on labels intertwine. The full closed-graph or source-resolvent construction in Proposition 11.6 also transports the actual unbounded domains.

For completeness, consider a bounded linear incidence A between two such reconstructed state interfaces. Its source-owned mixed matrix elements against the two dense labelled spans are retained. If U_1, U_2 are their reconstruction unitaries and A' is the corresponding incidence, these elements imply

$$\langle U_2 v, A' U_1 w \rangle = \langle v, A w \rangle \quad \text{for every labelled-span } v, w.$$

Density first gives $A' U_1 w = U_2 A w$ on that span and boundedness extends it to the full domain. For an unbounded incidence the retained labelled pairs are graph-norm dense in its full closed graph; equality of their joint data transports the closed graph itself. A shared formal differential expression would not suffice. The actual incoming/outgoing source maps use the same mixed data, so relative scattering attachments are also preserved.

Other native incidences are restrictions of their retained typed operation graphs, whose exact source descent has already been proved. In particular they are not inferred from equality of marginal orbit labels. Incoming and outgoing maps transform with the interface comparison and its inverse in their respective slots. Every actual sibling remains attached to the same joint operation. All generator squares consequently commute, and composition preserves their equations and exact pullback domains. The same argument on overlaps uses the retained chart transitions; it does not select new gluing maps or identify different lawful holonomies. $\square$

12.3 Complete role occupation over each response point

Theorem 12.4 (Target-fibred whole-interface role-occupancy closure). *For every $b \in \mathcal{B}_T$ on the received reconstructive domains, all seven role-occupancy completeness conditions hold for the whole-interface diagram of Definition 12.1. In particular, if $\mathcal{C}_b$ is the set of compatible complete eligible injections, its live residue is*

$$S_b^{\text{live}} = \bigcup_{a \in \mathcal{C}_b} \text{im } a = \{s_{b,i} : i \in I_T\},$$

where $s_{b,i}$ is the uniquely reconstructed lawful class for the source-required work i . Each compatible assignment is a bijection onto this residue, and the quotient of compatible assignments by lawful full-diagram automorphisms is a singleton. This result includes the full activated local/physical interface work, not only the quadratic support lattice.

Proof. Candidate completeness. The independent native reception classifies the target operations by their source, local-law, analytic and state/continuation types. Finite compositions are received with all mixed arguments; admitted analytic continuation, Hilbert completion and closed-operator constructions are covered by their separate reconstruction theorems. Thus every candidate for the stated work lies in one of the received interface families, not merely in a family selected by the desired support count.

Quotient completeness. At fixed elementary reconstruction data the local inverse, analytic reconstruction and joint Gram/closed-domain reconstruction provide actual source-preserving comparisons. Conversely each such comparison preserves the probes. Hence their lawful comparison relation is exactly the intrinsic reconstruction-data relation on each received interface. This classifies presentations of the whole interface; its identification with terminal observational identity requires the detectability result below.

Role completeness. The roles are the independently required whole-interface work of the source diagram. Repeated evaluations and its derived polarization or support objects introduce no extra interface burden. The reception includes every required generating incidence; a genuinely new operator or domain would change this target, not secretly add an occupant to its completed role list.

Eligibility completeness. Every generic candidate has an exact test against the specified source attachment and elementary reconstruction data. The reconstruction theorems show that an eligible candidate of role i has exactly the class $s_{b,i}$. Different typed roles have different interface types in the disjoint candidate union. No false injectivity assertion about distinct primitive particles is involved.

Compatibility completeness. Since $b \in \mathcal{B}_T$, an actual admitted native instance supplies one simultaneous assignment with all its incidence maps. Any other eligible compatible presentation has the same reconstructed objects and, by Lemma 12.3, the same whole-diagram class. All generator equations, including mixed and boundary equations, are tested; composite equations follow by composition. Rolewise eligibility alone was not used as a substitute for this argument.

Live-residue completeness. Examine any complete compatible injection. Its value at each i must be $s_{b,i}$. The assignment already supplied by the common native instance attains all these classes. The union of the images of all compatible injections is therefore exactly the displayed set. Each injection is onto that set, so the role-covering assignment condition follows after, not before, the live residue is formed. There is already one assignment at the lawful class level; passing to the smaller allowed full-tuple automorphism action leaves one orbit. No arbitrary graph symmetry is promoted to a lawful one.

Failure completeness. The received typed graphs and physical reconstructions retain admission domains, defined failures, zeros, obstructions, conditional values and undefined operations distinctly. A failed required attachment or missing physical domain cannot produce an eligible positive occupant. An inactive physical branch is absent from the activated role diagram, while a target requiring it without its admission witness is not certified by this theorem. These are the seven conditions of role-occupancy closure [13, Def. 7.1 and Thms. 7.2–7.4]. $\square$

12.4 When whole-interface reconstruction is terminal

Proposition 12.5 (Detectability of the reconstructive packet). *On a received target with the following actual readout interfaces, equality of the complete native observations is equivalent to equality of the intrinsic reconstructive packet:*

1. *the local interface uses the four/five-coordinate normalized observation family, or its whole intrinsic polynomial law is recoverable from admitted tensor evaluations;*
2. *the full labelled analytic-channel functions and their authorization, sheet and continuation statuses are admitted readouts;*

3. *the cyclic source interface has a fixed admitted invariant operator algebra closed under the products and adjoints used by its source words, with its vacuum correlation functions; its required closed-domain and incident-map data are recovered from actual native action/domain or mixed correlation probes.*

Only conditions for activated interfaces are needed. A source with fewer probes is classified by its reconstructed native readout quotient instead; no missing probe is inserted by this proposition.

Proof. Each normalized local invariant is an admitted quadratic, current, radial or vacuum-value observation by its construction. In the polynomial-law alternative, all required tensor coefficients follow from the admitted full-law evaluations and finite differentiation or polarization, under the retained source equivalence. A merely gauge-fixed raw coefficient would not meet this clause.

The full analytic function determines its convergent local germs by differentiation or Laurent expansion in the fixed channel frames. Its authorized domains and sheet/path statuses are separate native readouts, not inferred from the numerical function on an overlap. Thus its intrinsic analytic packet is detectable.

Write the source-generated vectors as $v_\lambda = A_\lambda \Omega$. The joint Gram entry is

$$K(\lambda, \mu) = \langle \Omega, A_\lambda^* A_\mu \Omega \rangle.$$

Under the stated native product/adjoint closure it is an admitted invariant correlation. No coherence between superselection sectors outside that algebra has been added. The joint kernel then reconstructs the canonical source-generated Hilbert space. The actual action/domain probes determine the graph there of each required closed operator, and graph-norm closure removes the presentation choice of a core. Mixed native correlations determine attached bounded maps on the dense labelled spans; the full-graph analogue handles admitted unbounded maps as in Lemma 12.3. All remaining native incidence entries are outputs of their admitted typed graph probes. Hence equality of native observations determines the entire intrinsic packet, without auxiliary witness names.

Conversely, packet equality reconstructs all these objects and their incident operations. Native factorization then preserves every admitted observation and continuation. The two kernels are equal. The source requirement is substantive: SYM-8 retains the admitted invariant correlation/response object, while the positive-metric invariant-field branch of SYM-2 supplies the vacuum/adjoint setting; neither licenses arbitrary charged cross-sector probes or untested closed domains [8, Def. 13.1][12, Sec. 8.2]. $\square$

Corollary 12.6 (Terminal family versus pointed realization). *The whole-interface realizations are classified over their actual intrinsic reconstruction image $\mathcal{B}_T$, with one lawful compatible presentation class over each reconstructed point. On the detectable native branches of Proposition 12.5, this image is also the terminal observational quotient and Theorem 12.4 is terminal role-occupancy closure. In general the terminal quotient is instead the image of the explicit reconstructed native readout map on $\mathcal{B}_T$. Distinct points separated by an admitted response observation remain distinct. In particular, a singleton assignment orbit over every point does not imply $|\mathcal{B}_T| = 1$, and it does not select a pole mass, coupling, state or boundary sector.*

Proof. The reconstruction and mixed-factorization results identify all presentations over a reconstructive point. When detectability holds, Proposition 12.5 identifies its kernel with the native observational kernel. Without that extra result, the explicit native readout evaluation still supplies the correct kernel by the native factorization theorem. A difference in an admitted readout obstructs terminal identification in either case. SCIC therefore gives exactly that observational quotient, not permission to delete a witnessed difference [14, Thms. 4.4 and 9.4]. Applying Theorem 12.4 fibrewise gives the stated presentation and, where detectable, terminal closure. $\square$

This closure does not promote a channel function, Hilbert space, interaction coefficient or interface map to a new primitive boson. Their necessity is the necessity of the work they perform at the declared target. The canonical photon, charged weak, neutral weak and radial Higgs supports remain the structural outputs already proved. A physical pole or asymptotic particle still requires its actual physical source domain; when present, its complete response is handled by the richer interface theorem rather than replaced by a count of those supports.

13 Minimal information and reconstructible descriptions

A complete response description need not be a minimal one. In particular, retaining a law, its differentiated vertices and a mass matrix does not make those three independent inputs. The same issue arises for physical channels that are already determined by retained state and operator data. We first remove such redundancy explicitly, then state the exact nonrepairable content of the terminal comparison.

Proposition 13.1 (Deletion of derived local descriptions). *On the normalized local target, retain the four or five invariant coordinates of Theorem 10.2 and all fixed independent sibling inputs. Stored copies of the local potential, gauge–Higgs vertices, quadratic response matrices, canonically computed supports and their projectors may be erased without loss of any admitted local observation. Their erasure requires no new source, parameter choice or repair.*

Proof. The explicit inverse reconstructs the same normalized gauge and scalar action. Its prescribed derivatives recover the vertices and quadratic forms. The retained kinetic and residual action determine the same canonical support projectors by the spectral and averaging constructions already proved. Thus every deleted entry is evaluated from retained information. Applying any admitted local observation after this reconstruction gives the original value on its original domain. $\square$

The proposition removes *copies* of data, not an independently varied amplitude on a support. The earlier amplitude-support minimality theorem and the local coordinate irredundancy theorem therefore remain different, compatible statements.

Theorem 13.2 (Resolvent-generated channel elimination). *Suppose an actual source certificate identifies an admitted channel as*

$$F_{uv}(z) = \langle u, (A - zI)^{-1}v \rangle$$

on its declared resolvent domain, where the complete state/operator interface reconstructs the closed operator A and its attached current vectors u, v . At a retained point z_0 of that domain write $R_0 = (A - z_0I)^{-1}$. Then

$$F_{uv}(z) = \sum_{n=0}^{\infty} (z - z_0)^n \langle u, R_0^{n+1}v \rangle \quad \text{when } |z - z_0| \|R_0\| < 1.$$

The channel germ is consequently a reconstructible duplicate and may be deleted while the certified operator/current interface and its domain/continuation authorization remain. The same conclusion holds for source-certified finite sums and compositions of such channels with retained maps and backgrounds.

Proof. On $\text{Dom } A$, put $t = z - z_0$. Then

$$A - zI = (I - tR_0)(A - z_0I).$$

If $\|t\| \|R_0\| < 1$, the Neumann series converges in bounded operator norm and yields

$$(A - zI)^{-1} = R_0(I - tR_0)^{-1} = \sum_{n=0}^{\infty} t^n R_0^{n+1}.$$

Taking the fixed current matrix element gives the displayed series. The reconstructed closed operator fixes its mathematical resolvent set, while the separately retained native authorization fixes which of those evaluations belong to the target. On a declared connected analytic continuation, the germ reconstruction theorem recovers the same continued channel and its retained sheet/path data. Finite certified compositions apply the same reconstruction to each constituent, with the original partial domains. No claim is made that an arbitrary channel is of this resolvent form without the actual source identification. $\square$

The full state interface can therefore be stronger than a list of analytic channels and also render that list redundant. Conversely, an independently supplied physical channel need not be reconstructible from a selected incomplete collection of state probes. A meromorphic background, an omitted cross-current pairing, or an unauthorized continuation cannot be supplied by appealing to a pole label.

Proposition 13.3 (Finite determination and its proper deletion target). *On the independently degree-bounded rational branch of Theorem 11.2, the retained finite jet reconstructs the full channel on the common domain. Stored higher coefficients are lossless deletions. On the general analytic branch there is no corresponding fixed finite truncation theorem.*

Proof. The rational reconstruction theorem identifies two admissible bounded degree rational functions from the stated finite jet, so every higher coefficient is determined. For an unrestricted holomorphic family, fix any proposed truncation through order m at z_0 . The functions 0 and $(z - z_0)^{m+1}$ have that same truncation and different admitted channel values. They are entire on the same marked domain. Thus finite-jet erasure fails even without poles or continuation ambiguity. This counterexample is within the holomorphic function family; it is not asserted to produce a second member of every narrower native physical image. $\square$

Theorem 13.4 (Exact irreducible terminal information). *Let $\mathcal{X}_T$ be the received native family, and let τ_T be its terminal invariant value obtained from the explicit native observation and continuation algebra. For any well-typed proposed retained datum $d(x)$, preserving the full target without new information is possible exactly when*

$$d(x) = d(y) \implies \tau_T(x) = \tau_T(y) \quad (x, y \in \mathcal{X}_T).$$

If a deletion fails, a primitive native readout after an admitted continuation separates a same-retained-data pair and rules out every reconstruction from those data. In particular, no proper quotient identifying two different terminal values is lossless. The terminal information object is unique up to the unique observation-preserving bijection, but its coordinate presentations and sufficient generating packets need not be unique or finite.

Proof. The native reconstruction and factorization theorem constructs the full target evaluation, including all partial-domain and failure values. Its terminal kernel is exactly equality of those native readouts after admitted continuations. Apply the semantic deletion criterion to that constructed evaluation. If the implication holds, assign to each retained value $d(x)$ the terminal value $\tau_T(x)$; this is well defined and reconstructs all target observations. If it fails, the complete native kernel theorem supplies an actual separating readout and continuation, so no function of the retained datum can recover both values.

If two information objects are each both sound and complete for these observations, their fibres on $\mathcal{X}_T$ coincide. Sending the value of one at x to the value of the other at x is therefore a well-defined bijection of their images, with the unique inverse preserving the same observations. A proper identification of two terminal values has the separating witness just described and cannot preserve those observations. This proves minimality as information, not minimality of a chosen array layout or a claim that an infinite family admits a finite irredundant coordinate basis. $\square$

The result makes three logically distinct minimality statements available. The normalized local family has an explicit irredundant four/five-coordinate chart and concrete deletion pairs. The analytic and state reconstructions identify recoverable functional data and expose failures of truncated or marginal packets. The full native terminal comparison then decides every proposed erasure on its *actual admitted image*. These results do not license a claim that every block of an enlarged packet is necessary merely because it was used in one proof.

Nor does nonprimitive status imply deletability. The spectral function, an interaction coefficient or a boundary restriction may be derivative of the same carrier structure and nevertheless essential to an admitted response comparison. Primitive carrier closure and target-relative information minimality answer different questions. Keeping that distinction permits a stronger physical atlas without multiplying primitive bosonic carriers.

14 The electroweak–Higgs realization atlas

The common source attachment links fixation geometry, interaction response and physical projection while preserving their different domains. Two theorems summarize the resulting atlas. The first gives the geometric, support and projection structure. The second establishes whole-response reconstruction, complete native comparison and target-relative closure, with observational detectability stated where it is needed.

Theorem 14.1 (Electroweak–Higgs realization and support classification). *On the independently source-admitted, dependency-expanded one-doublet branch, the following conclusions hold.*

1. *The native source determines a BCRS-I common assembly and its actual Higgs–electroweak attachment. The source-relative section preserves its fine input packets, boundary and continuation records. No universal premetric marked-response singleton is required.*
2. *Nonzero fixation determines a connected effective electromagnetic stabilizer line, a three-dimensional angular orbit tangent and a one-dimensional scalar normal quotient. The actual global stabilizer and source spectator records remain attached.*
3. *The gauge-response form is positive semidefinite of rank three, with kernel $C_g^{-1}(\mathfrak{h}_\phi)$. The residual action and kinetic form determine canonical electromagnetic, charged and neutral responsive modules. Every joint-invariant real gauge support is a sum of these modules.*
4. *With the scalar normal quotient separately typed, the invariant support lattice has four minimal nonzero elements. Their compatible module assignment has one lawful orbit. This result is a realization-support classification, not a count of primitive carriers or interacting particles.*
5. *In the declared renormalizable scalar target, the invariant potential is $a + bs + cs^2$. A nonzero stationary branch with positive radial Hessian has $c > 0$, $b < 0$, and one fixed-radius orbit of minima. More generally the radial response is controlled by $F''(s_0)$, with its positive, degenerate and negative branches retained.*

6. *On the licensed Minkowski realization, the null and timelike polarization quotients have complete positive fibres of dimensions two and three. The free helicity representation is constructed. An interacting photon identification is constructed on its independent spectral, Gram and asymptotic attachment domain.*
7. *On a fixed analytic continuation sheet, the regular Nielsen-type flow preserves determinant zeros and their multiplicities. An independently existing simple zero with nonzero physical numerator gives the actual pole, and the stated joint channel transport preserves its residue. These results do not create the spectral inputs they require.*
8. *Local radial-vector vertices and residual charge selection follow from the retained kinetic and representation data. Finite source-evaluated process diagrams compose on their exact common domains. On the positive quadratic branch, deleting any whole isotypic amplitude support loses an invariant observation and cannot be repaired by reconstruction from the remaining amplitudes.*

All conclusions commute with the lawful joint presentation comparisons on their domains. None changes the earlier BCRS-I carrier or response identity relation.

Proof. The source-relative attachment construction gives (i). The orbit, annihilator and real normal decomposition give (ii), with the inverse-image stabilizer retaining the actual global group. The Gram identity and coupled field/generator map give (iii). The exhaustive joint-invariant support theorem and the compatible assignment proof give (iv); in particular their count is taken after the subspaces have been determined. The invariant-polynomial and Hessian theorems give (v).

The polarization quotient, direct-integral helicity model and source-owned Gram extension prove (vi) on their respective domains. The determinant and external-channel transport equations prove (vii), retaining the independent noncancellation premise. The local kinetic expansion, character selection, process evaluation and amplitude erasure theorems prove (viii). Each construction was proved covariant with all incident maps retained. Conservative realization therefore preserves the predecessor object and its complete signature quotient throughout. $\square$

Theorem 14.2 (Native response reconstruction and exact closure). *On the received one-doublet target, the realization atlas has the following additional structure.*

1. *Its native local, channel, state and incidence operations form an explicit observation and continuation algebra, with exact fixed-sibling source descent, partial domains and failure values. Differential-polynomial laws are reconstructed from their finite coefficient tensors on the full declared base.*
2. *On the normalized positive local family, the complete local interaction response has a four-coordinate invariant normal form, or five coordinates when the vacuum value is observed. Its inverse reconstructs the entire stipulated gauge-Higgs action. The native local quotient is its actual admitted image.*
3. *Full analytic channel germs reconstruct the admitted meromorphic functions on their common marked connected domains. Independently bounded rational channels admit a finite determining jet. Joint source-word kernels reconstruct cyclic states, bounded actions and attached ports simultaneously; supplied resolvents or full closed graphs additionally reconstruct the exact unbounded operator domains.*
4. *Over each actual intrinsic reconstruction point, the entire activated interface diagram has complete eligibility, compatibility, live residue and one lawful assignment orbit. Where the reconstruction data are detectable by the native observations, this is full terminal role-occupancy closure. On a coarser observation target, its explicit readout quotient, rather than the unquotiented reconstruction record, is terminal.*

5. *The local invariant coordinates are irredundant on their declared open family. Derived local tensors and certified resolvent-generated channel germs are recoverable descriptions. On the full native target, a deletion is lossless exactly when it preserves the constructed terminal information value; otherwise an admitted readout after a continuation separates a same-retained-data pair and excludes every no-new-information reconstruction.*

These conclusions preserve distinct response parameters, physical branches and source domains. They require neither a universal singleton of response values nor a new primitive bosonic carrier.

Proof. Native coverage, polynomial reconstruction, certified relative operation descent and native factorization prove the first clause. The explicit inverse, local completeness and admitted-image theorems prove the second. Analytic-germ, bounded-rational, joint-kernel and closed-operator reconstruction prove the third on their separately specified domains.

The whole-diagram reconstruction lemma proves preservation of the generating mixed incidences, not merely of the marginal objects. The seven-part role-occupancy theorem then gives the fourth clause; the detectability proposition and native readout kernel distinguish terminal closure from a finer reconstruction presentation. The local deletion pairs, constructive duplicate eliminations and exact terminal-information theorem give the fifth. All are statements over the independently admitted source target and its actual image, so none selects a numerical response point or supplies an absent physical domain. $\square$

14.1 Exact branches and their interpretation

Domain	Established object	Distinction retained
Native one-doublet source	Common attachment and geometric descent	Not a lift of every formal premetric instance
Nonzero fixation and couplings	Residual carrier, rank-three response, complete invariant support lattice	Not a physical pole count
Positive radial Hessian	Stable radial quadratic response and support-minimal amplitude bundle	Not an interacting Higgs width
Zero radial Hessian	Degenerate radial quadratic response	Not a stable massive scalar claim
Negative radial Hessian	Unstable stationary branch at quadratic order	Not a metastable lifetime calculation
Licensed Minkowski envelope	Full kinematic polarization fibres	Not an interacting asymptotic-state construction
Free helicity branch	Explicit stable massless free representation	Not a substitute for the coupled source attachment
Source-owned photon attachment	Intertwining photon projection and its supplied asymptotic domain	Requires independent spectral, Gram and asymptotic data
Admitted physical pole channel	Existing pole and lawful analytic transport, with nonzero residue	Requires independent operator, continuation and pole-existence data

The richer response classifications have their own precise domains. Their information need not be finite, even when the primitive carrier structure and quadratic support lattice are finite.

Response comparison	Complete description and scope
Normalized local interaction	Four invariant coordinates, or five with an observed vacuum value; explicit inverse on the positive family and classification on the actual native image
Declared polynomial law	Finite jet tensors as sections on the full base; coefficient frames compared through their source maps
Admitted analytic channels	Full convergent germs on marked connected domains, with separate authorization and continuation data
Bounded rational channels	A finite determining jet; the degree bound is established independently by the response law
Cyclic state and continuation interface	One joint source kernel; actual resolvents or full closed graphs retain unbounded domains
Whole reconstructive interface	Complete compatible assignment over each intrinsic response point, with all mixed incidences retained
Observation-detectable interface	Terminal role-occupancy closure; a coarser observation target instead uses its native-readout quotient
Information deletion	Explicit local irredundancy, constructive removal of derived copies, and exact no-repair tests on the native image

The boundary cases delimit the continuation of the main theorem. At $\phi = 0$, the Higgs-induced response form has rank zero and the nonzero-orbit decomposition does not apply. If a coupling vanishes, C_g is no longer invertible. In canonical coordinates, $g_1 = 0$, $g_2 \neq 0$ can still give rank three, but the uncoupled Abelian field and the lost inverse coupling map prevent identifying this limit with the original target merely by its rank. For $g_2 = 0$, $g_1 \neq 0$, the form has rank one. These limits change the realization branch rather than its presentation.

The one-doublet support classification concerns internal gauge directions and the scalar normal quotient. Tensor-product and composite constructions retain their own source operations and must be classified at the corresponding operator or physical target. Extra scalar representations, changed global forms and new boundary sectors likewise require the appropriate source attachment and comparison. Additional field coordinates alone do not establish a new Higgs carrier identity.

14.2 Implications for further bosonic realization

Further bosonic realization can use the common-source attachment, geometric descent and complete quadratic support lattice together with the full local interaction normal form. The native operation algebra preserves the exact source, boundary and mixed-continuation information needed to combine this sector with other bosonic work. Analytic and state-space reconstruction determine the supplied physical responses, while the whole-interface theorem controls their simultaneous realization. Local irredundancy and the full native deletion test distinguish genuinely lost information from descriptions recoverable from the retained law.

These results provide the electroweak component of a common-boson atlas. Its interacting branches require more than response decomposition: the sources examined here do not independently construct the interacting photon asymptotic data or the full physical W , Z , and Higgs pole data. The corresponding projection theorems therefore retain those hypotheses. This is a limitation on what has been established from the sources, not a claim that the physical objects are absent. The bosonic construction also leaves the physical-realization boundaries of the preceding fermionic classification unchanged.

14.3 Conclusion

The one-doublet electroweak organization follows from a common fixation geometry. Its annihilator determines the residual charge direction, its coupled gauge action determines the massive quadratic directions, and its normal quotient carries the scalar response. The residual action and quadratic response together admit exactly three minimal real gauge supports. With the separately typed radial quotient, these determine a unique compatible four-module structure. This conclusion follows from an exhaustive classification of invariant supports, not from prescribing the familiar electroweak labels.

The same source structure supports a substantially richer response classification. The explicit local inverse reconstructs the complete normalized interaction law, not only its mass matrix. Full channel germs and joint source kernels reconstruct the admitted functional and state-space responses, including their relative current maps and actual continuation domains. The resulting native observation algebra determines terminal identity without treating a particle name, pole list or marginal response as a complete physical object.

Whole-interface uniqueness is a statement over each intrinsic response point. It becomes terminal closure when that interface's data are detectable by the admitted observations; otherwise the native readout quotient gives the appropriate identity. Genuine parameter and boundary distinctions survive either construction. Information minimality likewise concerns what the target can distinguish, not how many arrays a presentation stores. The local coordinate witnesses and resolvent reconstruction make that distinction explicit.

This yields a constraint-derived electroweak–Higgs representation with a complete structural core, reconstructive local and physical interfaces, and exact branchwise closure. Kinematic polarization, asymptotic photon realization and unstable resonance remain separately typed. Their common source relates these descriptions without turning them into new primitive carriers or replacing their independent physical domains.

Source dependencies

The source-first response objects and conservative realization rule are established in BCRS-I [1]. The kind-indexed fermionic comparison and its physical-realization boundaries are those of FCGS-III [2]. The standalone structural result supplies the shared Standard Model representation and bridge setting [3]. The Higgs bridge and metric-envelope papers distinguish the carrier roles and their order of dependence [4, 5]; gauge–vacuum compatibility and the residual normal form supply the preceding reduction data [6, 7].

SYM-8 supplies the principal electroweak realization results: the one-doublet target, stabilizer and response calculations, current transport, and regime-qualified Goldstone translation [8]. Its physical spectrum theorem retains independent spectral premises. The explicit finite factors and asymptotic remainder qualify the earlier longitudinal–Goldstone treatment [9]. SM-2, R7 and SYM-2 distinguish carrier standing, persistence-class identity, and stable versus resonant physical realizations [10, 11, 12]. Role-occupancy closure and standing carrier identity closure supply the assignment and full-signature criteria [13, 14]. These distinctions separate structural classification from physical pole existence. The geometric, polarization and analytic arguments particular to the present electroweak realization are proved in the text.

The complete native response comparison also uses the source-first symmetry transfer and the explicit source-incidence completion [17, 18]. Their use here preserves the particular admitted target and does not import another construction's role count. The numerical-standing and gauge-coupling results distinguish genuine invariant response data from changes of numerical presentation [16, 15]. The spectral-function witness makes the corresponding distinction between a carrier and its complete

functional response [19]. The local inverse, analytic and kernel reconstructions, and the mixed-interface classification are established in the present paper.

References

- [1] A. J. Maley, *Premetric Bosonic Response Structure in the Standard Model: Intrinsic Gauge Laws, Higgs Incidence, and Response Networks*. BCRS-I, Primitive-standing reference edition, 21 September 2026.
- [2] A. J. Maley, *Structural Universality and Exact Terminal Bifurcation: One-Higgs Naturalness, Projection Firewalls, and the Branch-Complete Fermionic Endpoint*. FCGS-III, Primitive-standing reference edition, 21 September 2026.
- [3] A. J. Maley, *The Standalone Standard Model Structure from First Principles: An AASC Derivation of Gauge-Product, Representation, Higgs/Bridge, Yukawa, Charge, Anomaly, Vacuum, and Parameter-Status Structure*. AASC manuscript (20 May 2026).
- [4] A. J. Maley, *The Higgs Bridge Carrier: Kernel Consequences for Electroweak Mass-Response*. Paper 1 of the Higgs Structural Endpoint Arc, AASC manuscript (June 2026).
- [5] A. J. Maley, *The Higgs Bridge and the Metric Envelope: An AASC Carrier-Order Witness for Hierarchy-Type Comparison*. AASC manuscript (15 June 2026).
- [6] A. J. Maley, *Gauge-Vacuum Compatibility: Vacuum Standing, Symmetry-Breaking Admissibility, and the Bridge to Mass-Generation Rigidity*. Gauge Realization Rigidity Arc, Paper A6, AASC manuscript (May 2026).
- [7] A. J. Maley, *Vacuum-Compatible Residual Sector Normal Form: Residual Gauge Structure after Admissible Vacuum-Mediated Reduction*. Pre-Standard-Model Realization Specialization Arc, Paper PSM6, AASC manuscript (May 2026).
- [8] A. J. Maley, *Electroweak Vacuum Fixation and Residual-Spectrum Rigidity: Gauge Quotient, Electromagnetic Stabilizer, Goldstone Translation, and the One-Doublet Electroweak Normal Form*. SYM-8, publication edition v1.3.0, AASC manuscript (9 August 2026).
- [9] A. J. Maley, *Equivalence Completion for Longitudinal Gauge Modes and Goldstone Degrees*. AASC manuscript, undated.
- [10] A. J. Maley, *Gauge Carriers: Photon, Gluon, and Weak-Boson Standing: Field Notation, Gauge Redescription, and No Primitive Gauge-Boson Skin Escape in the Standard Model*. SM-2 (July 2026). DOI: 10.5281/zenodo.21213644.
- [11] A. J. Maley, *Particles as Localized Persistence Classes: The Minimal Object Role of Particle Identity under Field-Carrier Mediation*. Realized Physics Arc, Paper R7, AASC manuscript (May 2026).
- [12] A. J. Maley, *Spacetime Symmetry as Domain Anchor: Poincaré, Lorentz, Translation, and CPT*. Kernel-Faithful, Constructor-Saturated, Semantically Causal, Projection-Recompositional, Apex-Closed Normal-Form Closure edition, SYM-2, AASC manuscript (15 July 2026).
- [13] A. J. Maley, *Role-Occupancy Closure under Admissibility: Why Role-Cardinality Matching Is Only the Cardinality Shadow of Compatible Realization*. AASC manuscript (May 2026).

- [14] A. J. Maley, *Standing Carrier Identity Closure: A Same-Scope Exhaustion Theorem for Source-Distinct Physical Realizations under AASC*. AASC manuscript (August 2026).
- [15] A. J. Maley, *Gauge-Coupling Parameters and Forced Uniqueness under Admissible Redescription*. AASC manuscript (2 May 2026).
- [16] A. J. Maley, *Deformation-Response and Numerical Standing over the SR7 Skeleton*. Mass-Generation and Numerical-Readiness Rigidity Arc, Paper MGN1, AASC manuscript (May 2026).
- [17] A. J. Maley, *SYM-11 Clean-Room Source-First Closure Derivation: Candidate Totality, Physical Occupancy, and Relative Support Removal*. Version 2.10.0, AASC manuscript (13 August 2026).
- [18] A. J. Maley, *Exact Certificate-Fibre Exhaustion and Unique Role-Occupancy Closure under AASC: A Target-Fibred Eight-Sorted Completion Theorem for the Constraint-Native Gravity-Quantum Capstone*. Module AASC-GRQM-MD-OCC-1.0, release v1.4.0-cosieve-support-corrected, AASC manuscript (2026).
- [19] A. J. Maley, *Structura Ex Necessitate Standardis Modelis: A Relative Closure Theorem for the Standard-Model Bookkeeping Interior under Fixed Load, Fixed Domain, and Certificate-Maximal Same-Domain Transport*. AASC manuscript (29 April 2026).